

Chapter 10 :

Single-Stage Small-Signal Amplifiers

10.1 Introduction

So far, we have discussed the AC analysis of single-stage amplifiers using BJTs and FETs. But we have not covered the design of single-stage amplifiers. Moreover, multistage amplifiers have not been studied yet. So, in this chapter the design and further discussion of single-stage small-signal amplifiers and multistage small-signal amplifiers will be covered. Small-signal amplifiers accept very small AC signals (small peak-to-peak value) in the input, and produce an amplified signal at the output without changing the frequency and shape of the signals. For small-signal amplifiers, the output is not more than, generally, ± 1 V. Sometimes, this output voltage is not sufficient to drive the output-stages. In that case, we have to use multi-stage amplifiers. Multi-stage amplifiers are constructed by cascading two or more single-stage amplifiers. The input signal is amplified by every stage. So, a very large signal is produced at the final stage. In this chapter different types of single-stage and multi-stage amplifiers with BJTs and FETs will be covered.

10.2 Single-Stage Common-Emitter Amplifier

Common-emitter amplifier is mostly used in electronic circuits. In Chapter 5, we have found that CE amplifier has reasonably high voltage gain, moderate current gain, medium input impedance, and high output impedance. As small-signal amplifier (also called voltage amplifier) CE configuration is a good choice. The operation of CE BJT amplifier has been discussed in Chapter 5. In addition to voltage amplification this amplifier produces 180° phase difference between the input and output signals. In some applications, e.g. oscillators, this phase-shift property is useful. However, if we do not need phase-shift, we can eliminate it by cascading even number of CE amplifier stages.

We can recapture the operation of this amplifier with the help of Fig.10-1. This figure shows the changes of voltage levels at different points of the amplifier. The input signal (v_s) is considered as a sinusoidal AC signal with very small peak-to-peak value. When this signal passes through the input-coupling capacitor (C_{ic}), it superimposes with the base-voltage V_B . So the base-voltage changes in-phase with the input signal. When the input signal increases, the base-voltage also increases, and hence, the base current increases. Because of the increase in base-current, collector-current also increases ($\because I_C = \beta I_B$). When the collector-current increases, the voltage-drop across the collector-resistor (R_C) increases. Therefore, the voltage level at the collector terminal (that was initially V_{CQ}) decreases. As the output is taken from the collector, the output voltage decreases. The reverse process occurs when the input signal decreases. As a small change in base current produces a very large change in collector current, an amplified voltage is produced at the collector. The collector-voltage swings in the same frequency and shape of the input signal but 180° out of phase. The Q-point of the amplifier oscillates with

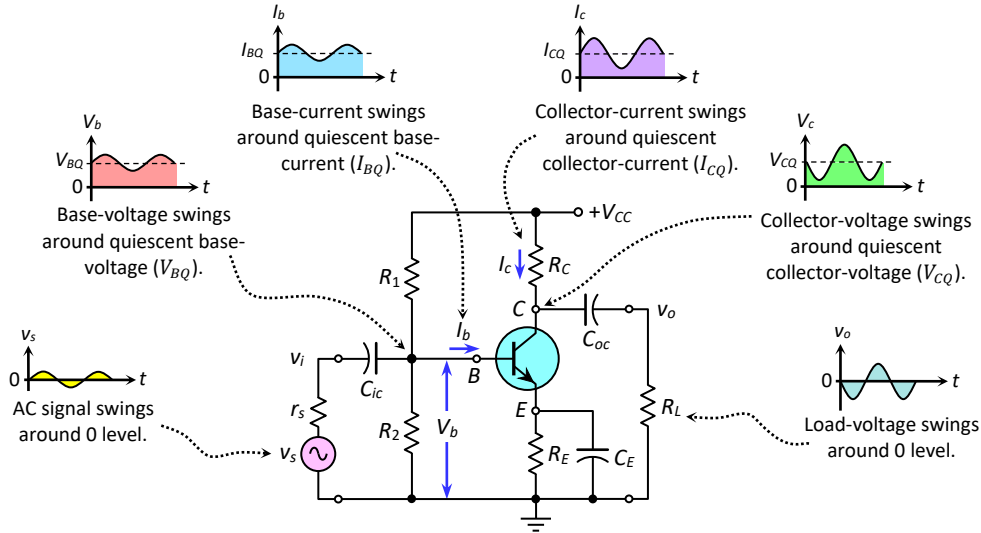


Fig.10-1: Waveforms of current and voltage swings at different points of CE amplifier

the input signal as described in Section 4.5. The amplified signal at the collector terminal has a DC voltage level as shown in Fig.10-1. When this signal passes through the output-coupling capacitor (C_{oc}), the capacitor blocks the DC voltage and the load resistor gets the pure amplified AC signal.

10.2.1 Single-Stage Common-Emitter Amplifier Design

For designing a CE amplifier generally, the following specifications are given:

The supply voltage, minimum voltage gain, frequency range, signal source impedance, and the load impedance.

As voltage divider bias gives best stability of operating point, it is most commonly used. Unless otherwise specified, in this book, voltage divider bias circuit will be used to design a BJT amplifier.

Selection of I_C , R_C , and R_E :

The main target of a small signal amplifier is to provide voltage gain. The expression of voltage gain of a CE amplifier is [Equ.8-],

$$A_v = -\frac{h_{fe}(R_C \parallel R_L)}{h_{ie}} \quad (10-1)$$

First of all, to get higher voltage gain, we have to choose a transistor with higher value of h_{fe} . However, to select the transistor we have to consider the other parameters: like maximum collector-emitter voltage, maximum frequency range, power capacity etc. For a fixed transistor, the

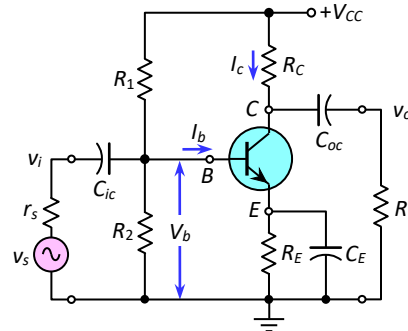


Fig.10-2: CE amplifier with voltage divider bias

gain will be high for higher value of $R_C \parallel R_L$. Larger value of R_C results in smaller voltage drop in R_E . In Chapter 4, we have found that higher voltage across the emitter resistance gives better stability. Here, a reasonable emitter voltage selection method is to consider $V_E = \frac{1}{4} V_{CC}$. Another approach is to set $V_E = 3 \text{ V}$ when $V_{CC} < 10 \text{ V}$ and $V_E = 5 \text{ V}$ for $V_{CC} \geq 10 \text{ V}$.

In designing the amplifier, we have to set quiescent collector current (I_C) large enough, so that the AC signal swing does not drive the transistor to saturation or cut-off region. I_C should be a little larger than the positive or negative swing produced by the AC signal. In case of small signal amplifier, we can set the quiescent collector current between 1 mA to 5 mA. R_C should be as large as possible, to maximize the voltage gain. However, larger R_C than R_L has less contribution to increase voltage gain, as A_v depends on $R_C \parallel R_L$ not only R_C . Moreover, to make the voltage gain of the amplifier independent of load resistance, sometimes we set R_C smaller than R_L . In general, $R_C = R_L$ is a good choice.

Once the collector current, and emitter voltage are set, the value of R_C will also be limited by the amount of voltage across the collector resistor, V_{RC} . To determine the value of V_{RC} , we have to decide the minimum value of V_{CE} .

Like the collector current, we have to set quiescent collector voltage (V_{CEQ}) large enough, so that the AC signal swing does not drive the transistor to saturation or cut-off region. It should be little larger than the positive or negative swing produced by the AC signal. We can set the quiescent collector voltage 3 V for $\pm 1 \text{ V}$ AC swing which is true for small signal amplifier. If AC swing is more than $\pm 1 \text{ V}$, V_{CE} should be greater than 3 V. Remember that, without driving the transistor into the saturation and cut-off region, minimum value of V_{CEQ} gives larger V_{RC} and hence larger R_C . In some cases, if V_{CC} is very large, the operating point of the transistor is set at the middle of the load line, that is, $V_{CE} = V_{CC}/2$.

Once the values of V_E , V_{RC} , and I_C are determined, R_C and R_E can be calculated using simply Ohm's law.

Bias Resistors:

Once the emitter voltage is set, the voltage at the base terminal will be $V_B = V_E + 0.7 \text{ V}$. The voltage drop across R_1 , and R_2 will be V_B , and $V_{CC} - V_B$. We know that,

$$\frac{R_1}{R_2} = \frac{V_{R1}}{V_{R2}} = \frac{V_{CC} - V_B}{V_B} \quad (10-2)$$

From this equation it is clear that, we can set different combinations of R_1 , and R_2 . But the values of R_1 , and R_2 should be high to provide a large input impedance, but their values should be low to provide high bias stability. As the voltage drops across the resistors are known, we can calculate their value by determining the current through them. A larger current will result in lower values of R_1 , and R_2 , while a smaller current will result in higher values. A good trade-off is to set the bias current one-tenth of the quiescent collector current, that is, $I_2 = I_{CQ}/10$. As we have seen in Chapter 5, for CE configuration input impedance mainly depends on h_{ie} . Therefore, for those amplifier (input impedance does not depend on biasing resistors), we can also set $I_2 = I_C/2$ to achieve better stability. But remember that, too much current in the biasing network will produce wastage of source power. For the circuits driven by small power source (like pencil batteries in remote controller, digital clock, etc.), small amount of power saving is also significant.

Bypass and Coupling Capacitors:

In AC analysis (Chapter 5), we have considered all coupling capacitors, and bypass capacitors short, but practically they provide some reactance and the lower cut-off frequency of the amplifier is imposed by them. In a single-stage CE amplifier, the bypass capacitor should have larger value than the coupling capacitors. Therefore, the lower cut-off frequency is mainly set by the bypass capacitor (C_E). The upper cut-off frequency is determined by the device parasitic capacitances. To keep the circuit size small, we have to select capacitors with minimum capacitance but large enough to provide the required lower cut-off frequency. That is why it is necessary to calculate the exact value of the capacitances for a given lower cut-off frequency.

In Chapter 5, we have derived the voltage gain of a CE amplifier with unbypassed emitter resistance. The expression is,

$$A_v = -\frac{h_{fe}(R_C \parallel R_L)}{h_{ie} + (1 + h_{fe})R_E} \quad (10-3)$$

Now, to calculate the value of C_E , it will not be considered as short-circuit. We have to consider the reactance of the bypass capacitor (C_E). Therefore, R_E of Equ.(10-3) will be $R_E \parallel X_{CE}$. So, Equ.(10-3) can be written as,

$$A_v = -\frac{h_{fe}(R_C \parallel R_L)}{h_{ie} + (1 + h_{fe})R_E \parallel X_{CE}}$$

Normally $R_E \gg X_{CE}$, so R_E can be omitted. Therefore,

$$A_v = -\frac{h_{fe}(R_C \parallel R_L)}{h_{ie} + (1 + h_{fe})X_{CE}} \quad (10-4)$$

As X_{CE} is a capacitive reactance, the denominator of Equ.(10-4) is a complex number, h_{ie} is real and $(1 + h_{fe})X_{CE}$ is imaginary. The magnitude of the voltage gain will be,

$$|A_v| = \frac{h_{fe}(R_C \parallel R_L)}{\sqrt{h_{ie}^2 + [(1 + h_{fe})X_{CE}]^2}} \quad (10-5)$$

At mid-frequency range (or flat-gain range), the reactance of C_E will be almost zero. Therefore, putting $X_{CE} = 0$ in Equ.(10-5), the maximum or mid-frequency voltage gain will be,

$$|A_{v(mid)}| = \frac{h_{fe}(R_C \parallel R_L)}{\sqrt{h_{ie}^2 + 0}} = \frac{h_{fe}(R_C \parallel R_L)}{h_{ie}}$$

At lower frequencies, X_{CE} will increase so the gain of the amplifier will decrease [Equ.(10-5)]. At the lower cut-off frequency (also called 3 dB point or half-power point) the real part will be equal to the imaginary part, i.e., $h_{ie} = (1 + h_{fe})X_{CE}$. Now, from Equ.(10-5),

$$A_v = \frac{h_{fe}(R_C \parallel R_L)}{h_{ie}\sqrt{1^2 + 1^2}} = \frac{h_{fe}(R_C \parallel R_L)}{h_{ie}\sqrt{2}} = \frac{A_{v(mid)}}{\sqrt{2}}$$

Therefore, at lower cutoff frequency (f_L),

$$h_{ie} = (1 + h_{fe})X_{CE}$$

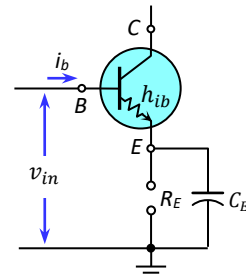


Fig.10-3: Series resistance seen by C_E .

$$\text{or,} \quad X_{CE} = \frac{h_{ie}}{1+h_{fe}} = h_{ib}$$

$$\text{or,} \quad \frac{1}{2\pi f_L C_E} = \frac{h_{ie}}{1+h_{fe}} = h_{ib} \quad (10-6)$$

$$\text{or,} \quad C_E = \frac{1+h_{fe}}{2\pi f_L h_{ie}} = \frac{1}{2\pi f_L h_{ib}} \quad (10-7)$$

The input signal (v_{in}) will be divided between h_{ib} and X_{CE} . The voltage drop across h_{ib} will be amplified, but the voltage drop across X_{CE} is just loss. Hence, the lower value of X_{CE} will give higher voltage gain.

Fig.10-3 shows the series resistance (h_{ib}) seen by the bypass capacitor.

To determine the values of the coupling capacitors, let us represent an amplifier by its equivalent circuit (Fig.10-4). If the reactances of the coupling capacitors become very small (theoretically zero), the amplifier will produce maximum voltage gain. But, if the reactances increase, some voltage will be dropped across them, and the gain of the amplifier will decrease. In case of CE amplifier, the cut-off frequency is determined by the bypass capacitor. If we calculate the values of the coupling capacitors considering 3 dB fall of gain, the overall fall of the gain at the cut-off frequency will be very high (9 dB). For this reason, we have to select coupling capacitors values such that their reactances become insignificant compared to the series resistance connected with them. A good approximation is to set the reactances of the coupling capacitors 10 times lower than the series resistance connected with them. So, the voltage drop across them will be less than 10%.

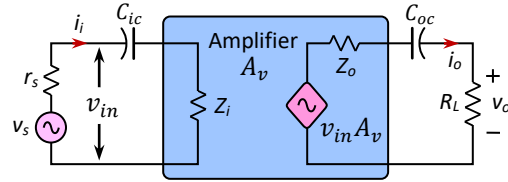


Fig.10-4: Series resistances seen by C_{ic} and C_{oc}

Input Coupling Capacitor:

From Fig.10-4 we find that, the resistance connected in series with C_{ic} are the source-resistance (r_s) and the input impedance of the amplifier (Z_i). Therefore, the reactance of C_{ic} should be 10 times smaller than $r_s + Z_i$ at the lower cut-off frequency. Hence,

$$X_{C_{ic}} = \frac{r_s + Z_i}{10} \text{ at } f_L$$

$$\text{or,} \quad \frac{1}{2\pi f_L C_{ic}} = \frac{r_s + Z_i}{10}$$

$$\therefore C_{ic} = \frac{10}{2\pi f_L (r_s + Z_i)}$$

For CE amplifier, $Z_i \approx h_{ie}$.

$$\therefore C_{ic} = \frac{10}{2\pi f_L (r_s + h_{ie})} \quad (10-8)$$

Output Coupling Capacitor:

From the Fig.10-4, we find that the resistance connected in series with C_{oc} are the amplifier output impedance (Z_o), and the load resistance (R_L). Therefore, the reactance of C_{oc} should be 10 times smaller than $Z_o + R_L$ at the lower cut-off frequency. Hence,

$$X_{C_{oc}} = \frac{Z_o + R_L}{10} \text{ at } f_L$$

or,

$$\frac{1}{2\pi f_L C_{oc}} = \frac{Z_o + R_L}{10}$$

$$\therefore C_{oc} = \frac{10}{2\pi f_L (Z_o + R_L)} \quad (10-9)$$

For CE amplifier, $Z_o \approx R_C$.

$$\therefore \boxed{C_{oc} = \frac{10}{2\pi f_L (R_C + R_L)}} \quad (10-10)$$

Another approach to calculate the input coupling capacitor is to make $X_{C_{ic}} = Z_i$ at two octaves below f_L , that is

$$X_{C_{ic}} = Z_i \text{ at } f_L/16$$

or,

$$\frac{1}{2\pi (f_L/16) C_{ic}} = Z_i$$

or,

$$\frac{1}{2\pi (f_L/16) C_{ic}} = Z_i$$

$$\therefore \boxed{C_{ic} = \frac{16}{2\pi f_L h_{ie}}} \quad (10-11)$$

Similarly, to make $X_{C_{oc}} = R_L$ at two octaves below f_L , that is

$$\boxed{C_{oc} = \frac{16}{2\pi f_L R_L}} \quad (10-12)$$

Example 10-1

Design a small-signal CE amplifier for the following specifications:

$V_{CC} = 20 \text{ V}$, $r_s = 600 \Omega$, $R_L = 3.3 \text{ k}$, lower cutoff frequency, $f_L = 200 \text{ Hz}$, transistor $h_{fe} = 50$, and $h_{ie} = 1 \text{ k}$.

Solution:

Let, $V_E = V_{CC}/4 = 5 \text{ V}$, $V_{CE} = 5 \text{ V}$ and $I_C = 2 \text{ mA}$.

$$\therefore R_E = \frac{5 \text{ V}}{2 \text{ mA}} = 2.5 \text{ k} \quad \text{use } 2.4 \text{ k standard, and } V_E = 4.8 \text{ V}$$

$$V_{RC} = 20 \text{ V} - 4.8 \text{ V} - 5 \text{ V} = 10.2 \text{ V}$$

$$\therefore R_C = \frac{10.2 \text{ V}}{2 \text{ mA}} = 5.1 \text{ k} \quad 5.1 \text{ k is a standard value.}$$

Let, $I_2 = 2 \text{ mA}/10 = 0.2 \text{ mA}$. The base voltage will be, $V_B = 4.8 \text{ V} + 0.7 \text{ V} = 5.5 \text{ V}$

$$\therefore R_2 = \frac{5.5 \text{ V}}{0.2 \text{ mA}} = 27.5 \text{ k} \quad \text{use standard } 27 \text{ k, and } I_2 = 0.204 \text{ mA}$$

Base current of the transistor will $I_B = 2 \text{ mA}/h_{fe} = 2 \text{ mA}/50 = 0.04 \text{ mA}$

Therefore, current through R_1 (I_1) will be $0.204 \text{ mA} + 0.04 \text{ mA} = 0.244 \text{ mA}$

$$\therefore R_1 = \frac{V_{CC} - V_B}{I_1} = \frac{20 \text{ V} - 5.5 \text{ V}}{0.244 \text{ mA}} = 59.4 \text{ k} \quad \text{use standard } 62 \text{ k}$$

Input coupling capacitor:

$$\text{Using Equ.(10-8), } C_{ic} = \frac{10}{2\pi f_1(r_s + h_{ie})} = \frac{10}{2\pi \times 200(0.6 + 1) \times 10^3}$$

$$\therefore C_{ic} = 5 \text{ } \mu\text{F, use standard } 5.6 \text{ } \mu\text{F}$$

Output Coupling Capacitor:

$$\text{Using Equ.(10-10), } C_{oc} = \frac{10}{2\pi f_L(R_C + R_L)} = \frac{10}{2\pi \times 200(5.1 + 3.3) \times 10^3}$$

$$\therefore C_{oc} = 0.95 \text{ } \mu\text{F, use standard } 1 \text{ } \mu\text{F}$$

Bypass Capacitor:

$$\text{Using Equ.(10-7), } C_E = \frac{1 + h_{fe}}{2\pi f_L h_{ie}} = \frac{1 + 50}{2\pi \times 200 \text{ Hz} \times 10^3} = 40.6 \text{ } \mu\text{F}$$

$$\text{or, } C_E = 40.6 \text{ } \mu\text{F use } 47 \text{ } \mu\text{F}$$

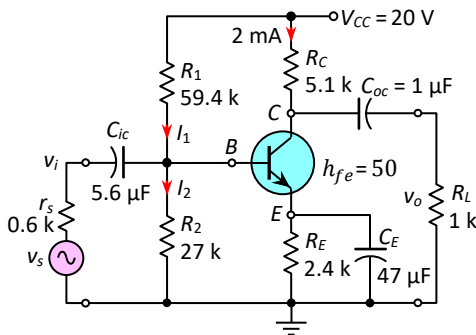


Fig.10-5: CE amplifier designed in Example 10.1

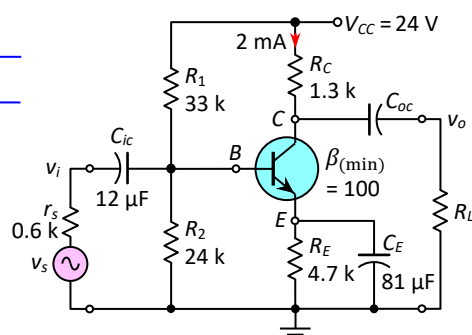


Fig.10-6: CE amplifier designed in Example 10.2

Results: The designed amplifier circuit is shown in Fig.10-5.

Comments: As the supply voltage is large, the design results in large R_C and hence large voltage gain. To achieve high voltage gain we could have used $I_C = 1 \text{ mA}$.

Example 10-2

Design a single stage CE amplifier with voltage divider bias that will give an open-circuit voltage gain of -100 . Assume, $V_{CC} = 24\text{ V}$, $I_{CQ} = 2\text{ mA}$, $f_1 = 100\text{ Hz}$, and $\beta_{min} = 100$.

Solution:

The operating point will be set at the middle of load line. To do that the value of V_{CE} should be half of V_{CC} and the value of saturation current should be double of quiescent current. Therefore,

Let,
$$V_{CE} = \frac{24\text{ V}}{2} = 12\text{ V}$$

and,
$$I_{C(sat)} = 2 \times I_{CQ} = 4\text{ mA}$$

The value of r_e will be,

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{2\text{ mA}} = 13\ \Omega$$

The value of R_C can be calculated from the expression of voltage gain. As the open-loop gain is given, R_L will be considered open. Using the voltage gain expression of r_e equivalent circuit of CE amplifier,

$$-\frac{R_C}{r_e} = -100 \text{ or, } R_C = 100 \times 13\ \Omega = 1.3\text{ k}$$

Value of R_E can be calculated from the saturation current as,

$$I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C + R_E}$$

or,
$$4\text{ mA} = \frac{24\text{ V} - 0.2\text{ V}}{1.3\text{ k} + R_E}$$

$\therefore R_E = 4.65\text{ k}$, use standard 4.7 k

The value of R_2 can be calculated from the following relation,

$$R_2 \leq \frac{\beta R_E}{10} \text{ or, } R_2 = \frac{50 \times 4.7\text{ k}}{10}$$

$$R_2 = \frac{100 \times 4.7\text{ k}}{10} = 23.5\text{ k, use standard } 24\text{ k}$$

Now, the emitter voltage is,

$$V_E = 2\text{ mA} \times 4.7\text{ k} = 9.4\text{ V}$$

$\therefore V_B = 9.4\text{ V} + 0.7\text{ V} = 10.1\text{ V}$

$\therefore R_1 = \frac{(V_{CC} - V_B) \times R_2}{V_B} = \frac{(24 - 10.1)\text{ V} \times 24\text{ k}}{10.1\text{ V}} = 33\text{ k}$

Bypass Capacitor:

Using Equ.(10-7),

$$C_E = \frac{1 + h_{fe}}{2\pi f_1 h_{ie}}$$

$\therefore$

$$C_E = \frac{1 + 50}{2\pi \times 100 \text{ Hz} \times 1 \times 10^3} = 81 \mu\text{F}$$

Input coupling capacitor:

Using Equ.(10-8),

$$C_{ic} = \frac{10}{2\pi f_1 (r_s + h_{ie})} = \frac{10}{2\pi \times 100 \text{ Hz} (0 + 1) \times 10^3} \approx 12 \mu\text{F}$$

As the value of R_L is not given, C_{oc} cannot be calculated.

Results: The designed amplifier circuit is shown in Fig.10-6.

Comments: Here, $R_E > R_C$. So more voltage will be dropped across R_E than across R_C . To get maximum voltage gain, we could have considered more voltage across R_C than R_E .

10.3 Single Stage Common-Base Amplifier

Although, common-emitter amplifiers are mostly used form of BJT amplifier configurations, in some cases we have to use common base amplifiers. Common-base amplifier can provide a reasonable level of voltage gain, but suffers from low input impedance and low current gain. In fact, the current gain of a CB amplifier is less than unity. However, these amplifiers are used in high-frequency applications like RF amplifiers for its better frequency response than CE amplifier. On the other hand, due to its very low input impedance, it is used to amplify signals from microphone. Due to the unity current gain, this amplifier is also used as **current buffer**. The operation process of CB amplifier is described in **Chapter** . Unlike the CE amplifier, CB amplifier does not produce phase difference between the input and output signal.

10.3.1 Single-Stage Common-Base Amplifier Design

Like CE amplifiers, for designing a CB amplifier generally, the following specifications are given:

The supply voltage, minimum voltage gain, frequency range, signal source impedance, and the load impedance.

The design technique is almost similar to that of the CE amplifier. Here also, the voltage divider bias is most commonly used.

Selection of I_C , R_C , and R_E :

The expression of voltage gain of a CB amplifier is [Equ.8-],

$$A_v = \frac{h_{fb}(R_C \parallel R_L)}{h_{ib}} = \frac{(R_C \parallel R_L)}{r_e} \quad (10-13)$$

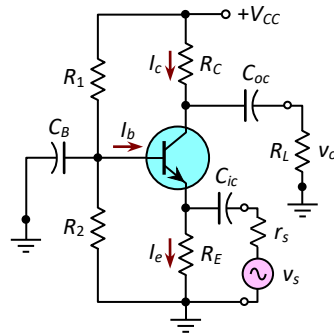


Fig.10-7: CB amplifier with voltage divider bias

As the value of h_{fb} is almost 1, the voltage gain of CB amplifier does not depend on h_{fb} . So, the transistor has to be selected considering the other parameters: like maximum collector-emitter voltage, maximum frequency range, power capacity etc. For a fixed transistor, the gain will be high for higher value of $R_C \parallel R_L$ and lower value of r_e . A larger emitter current will result in lower r_e , i.e., higher voltage gain. However for a fixed supply voltage larger emitter current may reduce the value of R_C . To achieve a desired voltage gain we have to select appropriate value of R_C . Larger value of R_C results in higher voltage gain but, making R_C much larger than R_L has little contribution in increasing voltage gain. Therefore, for a known value of R_L , a good choice is $R_C = R_L$. If the value of R_L is not known, to make the gain of the amplifier independent of R_L , sometimes we select lower value for R_C .

In CB amplifier, the emitter voltage will be set exactly in the same method described for CE amplifier. That is $V_E = \frac{1}{4}V_{CC}$ or, $V_E = 3\text{ V}$ when $V_{CC} < 10\text{ V}$ and $V_E = 5\text{ V}$ for $V_{CC} \geq 10\text{ V}$.

In designing the amplifier, we have set quiescent collector current (I_C) large enough, so that the AC signal swing does not drive the transistor to saturation or cut-off region. It should be little larger than the positive or negative swing produced by the AC signal. Like CE amplifier, we can set the quiescent collector current between 1 mA to 5 mA.

Like the collector current, we have set quiescent collector voltage (V_{CEQ}) large enough, so that the AC signal swing does not drive the transistor to saturation or cut-off region. It should be little larger than the positive or negative swing produced by the AC signal. As before, we can set the quiescent collector-emitter voltage 3 V to 5 V for small signal amplifiers. Without driving the transistor into the saturation and cut-off region by the AC signal, minimum value of V_{CEQ} gives larger V_{RC} and hence larger R_C . In some cases, if V_{CC} is very large, the operating point of the transistor is set at the middle of the load line considering, $V_{CE} = V_{CC}/2$.

Once the values of V_E , V_{RC} , and I_C are determined, R_C and R_E can be calculated using simply Ohm's law.

Bias Resistors:

Once the emitter voltage is set, the voltage at the base terminal can be calculated as $V_B = V_E + 0.7\text{ V}$. The voltage drop across R_1 , and R_2 will be V_B , and $V_{CC} - V_B$, respectively. In CE amplifier the input impedance depends on the biasing resistance, but in CB amplifier it does not. So can select lower values of R_1 and R_2 to achieve better stability. But too small values of these resistors, will cause wastage of power. A good choice is, to set the bias current one-tenth to half of the quiescent collector current, that is, $I_2 = \frac{1}{10}I_{CQ}$ to $\frac{1}{2}I_{CQ}$.

Bypass and Coupling Capacitors:

In Chapter 5, we have derived voltage gain of a CB amplifier with unbypassed base resistance. The expression is,

$$A_v = \frac{h_{fb}R'_L}{h_{ib} + (1 - h_{fb})R_B} \quad (10-14)$$

Now, to calculate the value of C_B , we have to consider its reactance. As shown in Fig.5-41, X_{CB} will come in parallel to R_B . Therefore,

$$A_v = \frac{h_{fb}R'_L}{h_{ib} + (1 - h_{fb})R_B \parallel X_{CB}}$$

Normally $R_B \gg X_{CB}$, so,

$$A_v = \frac{h_{fb}R'_L}{h_{ib} + (1 - h_{fb})X_{CB}} \quad (10-15)$$

Magnitude of the voltage gain will be,

$$|A_v| = \frac{h_{fb}R'_L}{\sqrt{h_{ib}^2 + [(1 - h_{fb})X_{CB}]^2}} \quad (10-16)$$

At mid-frequency range, the reactance of C_B will be almost zero. Therefore, putting $X_{CB} = 0$ in Equ.(10-16), the maximum or mid-frequency voltage gain will be,

$$|A_{v(mid)}| = \frac{h_{fb}R'_L}{\sqrt{h_{ib}^2 + 0}} = \frac{h_{fb}R'_L}{h_{ib}}$$

At lower cut-off frequency, $h_{ib} = (1 - h_{fb})X_{CB}$. Now, from Equ.(10-16),

$$|A_v| = \frac{h_{fb}R'_L}{h_{ib}\sqrt{1^2 + 1^2}} = \frac{h_{fb}R'_L}{h_{ib}\sqrt{2}} = \frac{A_{v(mid)}}{\sqrt{2}}$$

Therefore, at lower cut-off frequency, f_L

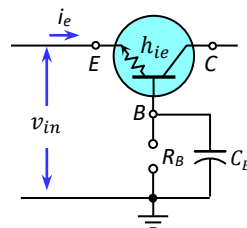


Fig.10-8: Series resistance seen by C_B

$$h_{ib} = (1 - h_{fb})X_{CB}$$

$$\therefore X_{CB} = \frac{h_{ib}}{1 - h_{fb}} = h_{ie}$$

$$\text{or,} \quad \frac{1}{2\pi f_L C_B} = \frac{h_{ib}}{1 - h_{fb}} = h_{ie}$$

$$\text{or,} \quad C_B = \frac{1 - h_{fb}}{2\pi f_L h_{ib}} = \frac{1}{2\pi f_L h_{ie}} \quad (10-17)$$

Fig.10-8 shows the series resistance and reactance seen by the input signal.

In CE amplifier the lower cut-off frequency is determined by the emitter bypass capacitor because it is the largest capacitor. But in common-base amplifier, the input capacitor is the largest. So, the lower cut-off frequency will be determined by the input capacitor not by the bypass capacitor. Therefore, the bypass capacitor will be considered 10 times higher than the calculated value by Equ.(10-17).

$$\therefore C_B = \frac{10(1 - h_{fb})}{2\pi f_L h_{ib}} = \frac{10}{2\pi f_L h_{ie}} \quad (10-18)$$

Input Coupling Capacitor:

As the input coupling capacitor is the largest capacitor in CB amplifier, its value should be calculated considering its reactance equal to the total series resistance in its path. From Fig.10-4, the resistance connected in series with C_{ic} are the source-resistance (r_s) and the input impedance of the amplifier (Z_i). Therefore,

$$X_{C_{ic}} = r_s + Z_i \text{ at } f_L$$

The input impedance of a CB amplifier is, $Z_i = h_{ib}$.

$$\therefore \frac{1}{2\pi f_L C_{ic}} = r_s + h_{ib}$$

$$\therefore C_{ic} = \frac{1}{2\pi f_L (r_s + h_{ib})} \quad (10-19)$$

Output Coupling Capacitor:

In Fig.10-4 we have seen that the resistance connected in series with C_{oc} are the amplifier output impedance (Z_o), and the load resistance (R_L). Therefore, the reactance of C_{oc} will be 10 times smaller than $Z_o + R_L$ at the lower cut-off frequency. Hence,

$$X_{C_{oc}} = \frac{Z_o + R_L}{10} \text{ at } f_L$$

$$\text{or, } \frac{1}{2\pi f_L C_{oc}} = \frac{Z_o + R_L}{10}$$

$$\therefore Z_o = R_C, \quad C_{oc} = \frac{10}{2\pi f_L (R_C + R_L)} \quad (10-20)$$

Example 10-3

Design a small signal CB amplifier for the following given specifications:

$V_{CC} = 12 \text{ V}$, $r_s = 600 \Omega$, $R_L = 4.7 \text{ k}$, lower cut-off frequency, $L = 200 \text{ Hz}$, transistor $h_{fe} = 100$.

Solution:

Let, $V_E = V_{CC}/4 = 3 \text{ V}$, $V_{CE} = 3 \text{ V}$ and $I_C = 2 \text{ mA}$.

$$\therefore R_E = \frac{3 \text{ V}}{2 \text{ mA}} = 1.5 \text{ k [Ans.]}$$

$$V_{RC} = 12 \text{ V} - 3 \text{ V} - 3 \text{ V} = 6 \text{ V}$$

$$\therefore R_C = \frac{6 \text{ V}}{2 \text{ mA}} = 3 \text{ k [Ans.]}$$

Let, $I_2 = 2 \text{ mA}/10 = 0.2 \text{ mA}$. The base voltage will be, $V_B = 3 \text{ V} + 0.7 \text{ V} = 3.7 \text{ V}$

$$\therefore R_2 = \frac{3.7 \text{ V}}{0.2 \text{ mA}} = 18.5 \text{ k} [\text{Ans.}] \quad \text{use standard } 18 \text{ k, and } I_2 = 0.206 \text{ mA}$$

Base current of the transistor will $I_B = 2 \text{ mA}/h_{fe} = 2 \text{ mA}/100 = 0.002 \text{ mA}$

Therefore, the current through R_1 will be $I_1 = 0.206 \text{ mA} + 0.002 \text{ mA} = 0.208 \text{ mA}$

$$\therefore R_1 = \frac{V_{CC} - V_B}{I_1} = \frac{12 \text{ V} - 3.7 \text{ V}}{0.208 \text{ mA}} = 40.1 \text{ k} [\text{Ans.}] \quad \text{use standard } 39 \text{ k} + 1.1 \text{ k}$$

Input coupling capacitor:

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{I_C} = \frac{26 \text{ mV}}{2 \text{ mA}} = 13 \Omega = h_{ib}$$

$$\therefore h_{ie} \approx h_{fe} h_{ib} = 100 \times 13 \Omega = 1.3 \text{ k}$$

Using Equ.(10-19),

$$C_{ic} = \frac{1}{2\pi f_L (r_s + h_{ib})} = \frac{1}{2\pi \times 200 (600 + 13)}$$

$$\therefore C_{ic} = 1.3 \mu\text{F}, \text{ use standard } 1.5 \mu\text{F} [\text{Ans.}]$$

Output coupling capacitor:

$$\text{Using Equ. (10-20)} \quad C_{oc} = \frac{10}{2\pi f_L (R_C + R_L)} = \frac{10}{2\pi \times 200 (3 + 4.7) \times 10^3}$$

$$\therefore C_{oc} = 1.1 \mu\text{F}, \text{ use standard } 1.5 \mu\text{F} [\text{Ans.}]$$

Bypass Capacitor:

$$\text{Using Equ. (10-18),} \quad C_B = \frac{10}{2\pi f_L h_{ie}} = \frac{10}{2\pi \times 200 \times 1.3 \text{ k}}$$

$$\text{or,} \quad C_B = 0.6 \mu\text{F} \text{ use } 1 \mu\text{F}$$

Results: The designed amplifier circuit is shown in Fig.10-9.

Comments: It is mentioned that, for CB amplifier the input coupling capacitor has the largest value. But here, C_{ic} and C_{oc} have the same values. The reason is that, C_{ic} has been calculated considering its reactance equal to the series resistance in its path, while the other capacitors have been calculated considering their reactances 10 times less than their series resistance.

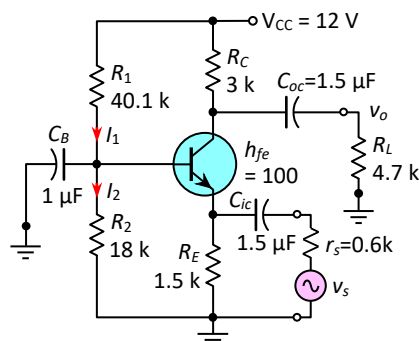


Fig.10-9: CB amplifier designed in Example 10.3

10.4 Single Stage Common-Collector Amplifier

Common-collector amplifiers can provide a very high current gain, but suffers from low voltage gain. In fact, the voltage gain of a CC amplifier is approximately unity. Other advantages of CC amplifier are: it has high input impedance and very low output impedance. Due to the unity voltage gain, CC amplifier is called *voltage follower* or *emitter follower*. The emitter follower configuration is mostly used as a voltage buffer, power amplifier, etc. These configurations are also used in impedance matching applications because of their high input impedance. The operation process of CB amplifier is described in Chapter . Like the CB amplifier, there is no phase difference between the input and output signal in CC configuration.

10.4.1 Single-Stage Common-Collector Amplifier Design

For designing a CB amplifier generally, the following specifications are given:

The supply voltage, minimum current gain, frequency range, signal source impedance, and the load impedance.

The design technique is almost similar to that of CE and CB amplifier. Here also, the voltage divider bias is most commonly used, but in some cases, simple base bias is also used. The circuit diagram of a CC amplifier with voltage divider bias is shown in Fig.10-10.

Selection of I_C , and R_E :

The expression for voltage gain of a CC amplifier is [Equ.8-],

$$A_v = \frac{R_E \parallel R_L}{r_e + R_E \parallel R_L} \quad (10-21)$$

and the expression for current gain of a CC amplifier is [Equ.8-],

$$A_i = \frac{i_o}{i_{in}} = \frac{(\beta + 1)R_B R_E}{(R_E + R_L)[R_B + \beta(r_e + R_E \parallel R_L)]} \quad (10-22)$$

From Equ.(10-21), we find that the voltage gain will be high for low values of r_e and high values of $R_E \parallel R_L$. To minimize the value of r_e , we have to set higher quiescent current I_{CQ} , because $r_e = 26 \text{ mV}/I_E \approx 26 \text{ mV}/I_{CQ}$. To increase the value of $R_E \parallel R_L$, we can use large R_E . From Equ.(10-22), we see that larger value of R_E will also increase current gain. Larger value of $R_E \parallel R_L$ will result in larger input impedance $[\beta(r_e + R_E \parallel R_L)]$, which decreases the current gain. This problem can be omitted by increasing the bias resistance R_B . That is, to get higher current gain, R_B should be greater than $\beta(r_e + R_E \parallel R_L)$ as much as possible.

As CC amplifier does not amplify voltage, the input signal of this amplifier has to be pre-amplified, using voltage amplifiers. Moreover, as this amplifier will amplify the current, the output signal will have both high voltage swing and current swing. To maximize the swing, the operating point of the amplifier has to be set at the middle of the load-line. In designing voltage amplifiers, we have considered $I_{CQ} = 1$ to 5 mA. But for CC amplifier, I_{CQ} should be at least equal to the peak value of

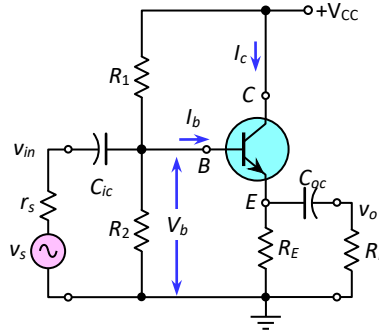


Fig.10-10: CC amplifier with voltage divider bias

output current swing. For example, if the AC output current flowing through $R_E \parallel R_L$ has an RMS value of 100 mA. The peak value (or amplitude) of this AC current will be $100 \times \sqrt{2} \approx 141$ mA. Therefore, I_{CQ} should be slightly (1 to 3 mA) greater than 141 mA. In fact, this current will be divided between R_E and R_L in proportion to their values. The transistor used in CC amplifier must have the capacity to tolerate this high amount of collector current.

As there is no collector resistor, equal distribution of V_{CC} across R_E and V_{CE} will set the operating point at the middle of load line.

Once the values of V_E , and I_C are determined, R_E can be calculated using Ohm's law.

Bias Resistors:

Once the emitter voltage is set, the voltage at the base terminal can be calculated as $V_B = V_E + 0.7$ V. The voltage drop across R_1 , and R_2 will be V_B , and $V_{CC} - V_B$, respectively. To make the current gain very high, the value of R_B should be greater than the input impedance at the base terminal $[\beta(r_e + R_E \parallel R_L)]$. However, very large value of R_B will sacrifice the stability of the amplifier. The value of R_B depends on R_1 , and R_2 . The value of R_1 , and R_2 can be determined if we know the current flowing through them. A good choice is, to set the current of R_2 one-tenth to half of the quiescent collector current, that is, $I_2 = \frac{1}{10} I_{CQ}$ to $\frac{1}{2} I_{CQ}$.

Output Coupling Capacitor:

In common-collector amplifier there are only two coupling capacitors, and no bypass capacitor. Among the two capacitors, generally the output coupling capacitor will have the larger value. Therefore, the lower cut-off frequency will be determined by the output coupling capacitor. Hence, in calculating its value, we will consider its reactance equal to the total series resistance in its path. From Fig.10-2, we find that the resistance connected in series with C_{oc} are the output impedance of the amplifier (Z_o), and the load resistance (R_L). Therefore,

$$X_{C_{oc}} = Z_o + R_L \text{ at } f_L$$

$$\text{or, } \frac{1}{2\pi f_L C_{oc}} = Z_o + R_L \quad (10-23)$$

For common-collector amplifier the output impedance is,

$$Z_o = [(r_s \parallel R_B) / \beta + r_e] \parallel R_E$$

$$\therefore \text{ from Equ.(10-23), } C_{oc} = \frac{1}{2\pi f_L [(r_s \parallel R_B) / \beta + r_e] \parallel R_E + R_L} \quad (10-24)$$

Input Coupling Capacitor:

In CC amplifier, the lower cut-off frequency is determined by the output coupling capacitor. So input coupling capacitor should be large enough so that its reactance becomes very small at the lower cut-off frequency. One rule is to consider its reactance one-tenth of the total series resistance connected with it. From Fig.10-2, the resistance connected in series with C_{ic} are the source resistance (r_s) and the input impedance of the amplifier (Z_i). Therefore,

$$X_{C_{ic}} = (r_s + Z_i) / 10 \text{ at } f_L$$

The input impedance of a CC amplifier is, $Z_i = R_B \parallel \beta(r_e + R_E \parallel R_L) \approx \beta(r_e + R_E \parallel R_L)$

$$\therefore \frac{1}{2\pi f_L C_{ic}} = [r_s + \beta(r_e + R_E \| R_L)]/10$$

$$\therefore C_{ic} = \frac{10}{2\pi f_L [r_s + \beta(r_e + R_E \| R_L)]} \quad (10-25)$$

Example 10-4

Design a small-signal CC amplifier for the following given specifications:

$V_{CC} = 12\text{ V}$, $r_s = 600\ \Omega$, $R_L = 100\ \Omega$, lower cut-off frequency, $f_L = 100\text{ Hz}$, minimum current gain of 40, and transistor $h_{fe} = 100$.

Solution:

To set the DC operating point at the middle of the load line,

Let,
$$V_{CE} = V_E = \frac{1}{2} V_{CC} = \frac{1}{2} \times 12\text{ V} = 6\text{ V}$$

The larger value of emitter resistance will give better current gain. So let,

$$R_E = 10R_L = 10 \times 100\ \Omega = 1\text{ k}$$

$$\therefore I_E = I_{CQ} = \frac{V_E}{R_E} = \frac{6\text{ V}}{1\text{ k}} = 6\text{ mA}$$

Using Equ.()
$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{6\text{ mA}} \approx 4.3\ \Omega$$

As the current gain of the amplifier should be 40, so

$$A_i = \frac{(\beta + 1)R_B R_E}{(R_E + R_L)[R_B + \beta(r_e + R_E \| R_L)]} = 40$$

Now, solving this equation, we get $R_B = 7.4\text{ k}$

As $V_E = 6\text{ V}$, the base voltage will be,

$$V_B = V_E + 0.7\text{ V} = 6\text{ V} + 0.7\text{ V} = 6.7\text{ V} = V_{R1}$$

Therefore,
$$V_{R1} = V_{CC} - V_{R1} = 12\text{ V} - 6.7\text{ V} = 5.3\text{ V}$$

Now we can relate R_1 , and R_2 as

$$\frac{R_1}{R_2} = \frac{V_{R1}}{V_{R2}} = \frac{5.3\text{ V}}{6.7\text{ V}}$$

or,
$$R_1 = 0.79 R_2$$

But we need
$$R_B = \frac{R_1 R_2}{R_1 + R_2} = 7.4\text{ k}$$

$$\text{or,} \quad \frac{0.79 R_2 R_2}{0.79 R_2 + R_2} = 7.4 \text{ k}$$

$$\text{or,} \quad \frac{0.79 R_2}{1.79} = 7.4 \text{ k}$$

$$\therefore R_2 = \frac{7.4 \text{ k} \times 1.79}{0.79} \approx 16 \text{ k}$$

$$\therefore R_1 = 0.79 \times 16 \text{ k} \approx 13 \text{ k}$$

Input coupling capacitor:

$$\text{Using Equ.(10-25),} \quad C_{ic} = \frac{10}{2\pi f_L [r_s + \beta(r_e + R_E || R_L)]}$$

$$\text{or,} \quad C_{ic} = \frac{10}{2\pi \times 100 [600 + 100(4.3 + 1000 || 100)]} \text{ F} = 1.6 \mu\text{F}$$

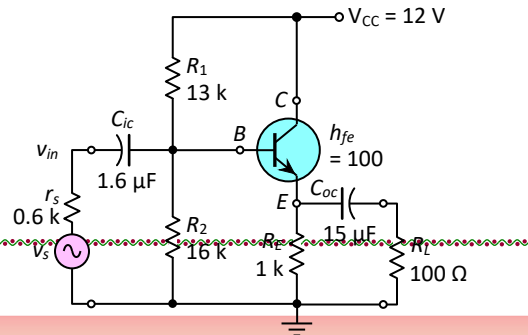
Output coupling capacitor:

$$\text{Using Equ.(10-24),} \quad C_{oc} = \frac{1}{2\pi f_L \{(r_s || R_B) / \beta + r_e\} || R_E + R_L}$$

$$\text{or,} \quad C_{oc} = \frac{1}{2\pi \times 100 \{[(0.6 \text{ k} || 7.4 \text{ k}) / 100 + 4.3] || 1000 + 100\}} \text{ F} \approx 15 \mu\text{F}$$

Results: The designed amplifier circuit is shown in Fig.10-9.

Comments: Here, the desired current gain is achieved by calculating the value of R_B for a fixed value of R_E . We can also do that by calculating the value of R_E for a fixed value of R_B .



Example 10-5

Design a small signal CC amplifier for the following given specifications:

$V_{CC} = 24 \text{ V}$, $r_s = 100 \Omega$, $R_L = 100 \Omega$, $f_L = 100 \text{ Hz}$, minimum current gain of 100, and transistor $h_{fe} = 100$.

Solution:

The current gain of a CC amplifier will always be less than the device current gain. In this example, the transistor current gain is 100, and the required current gain of the amplifier is also 100. So we cannot design the CC amplifier with these requirements.

Example 10-6

Design a small signal CC amplifier for the following given specifications:

Fig.10-11: CC amplifier designed in Example 10-5

$R_L = 50 \Omega$, lower cut-off frequency, $f_1 = 100 \text{ Hz}$, minimum RMS current through the load is 100 mA , and $h_{fe} = 100$.

Solution:

The peak current through the load will be, $100 \text{ mA} \times \sqrt{2} \approx 141 \text{ mA}$

let, $R_E = 10R_L = 10 \times 50 \Omega = 500 \Omega$

The peak voltage across the load will be $141 \text{ mA} \times 50 \Omega = 7.5 \text{ V}$

let, $V_E = 8 \text{ V}$ and $V_{CE} = 8 \text{ V}$

Equivalent load is $R'_L = 500 \Omega \parallel 50 \Omega = 45.5 \Omega$

The emitter current will be,

$$I_E = \frac{V_E}{R'_L} = \frac{8 \text{ V}}{45.5 \Omega} \approx 176 \text{ mA} = I_{CQ}$$

Value of V_{CC} , $V_{CC} = V_E + V_{CE} = 8 \text{ V} + 8 \text{ V} = 16 \text{ V}$

Base current will be,

$$I_B = \frac{176 \text{ mA}}{h_{fe}} = \frac{176 \text{ mA}}{100} = 1.76 \text{ mA}$$

Let, $I_2 = 2 \times I_B = 2 \times 1.76 \text{ mA} = 3.5 \text{ mA}$

$$\therefore R_2 = \frac{V_E + 0.7 \text{ V}}{I_2} = \frac{8 \text{ V} + 0.7 \text{ V}}{3.5 \text{ mA}} = 2.5 \text{ k (use 2.4 k)}$$

$$I_1 = I_2 + I_B = 3.5 \text{ mA} + 1.76 \text{ mA} = 5.26 \text{ mA}$$

$$\therefore R_1 = \frac{V_{CC} - (V_E + 0.7 \text{ V})}{I_1} = \frac{16 \text{ V} - (8 \text{ V} + 0.7 \text{ V})}{5.26 \text{ mA}} \approx 1.4 \text{ k}$$

$$\therefore R_B = R_1 \parallel R_2 = 1.4 \text{ k} \parallel 2.4 \text{ k} \approx 884 \Omega$$

Input coupling capacitor:

$$\text{Value of } r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{176 \text{ mA}} \approx 0.15 \Omega$$

$$\text{Using Equ. (10-23), } C_{ic} = \frac{1}{2\pi f_L [r_s + \beta(r_e + R_E \parallel R_L)]}$$

$$\text{or, } C_{ic} = \frac{1}{2\pi \times 100 [600 + 100(0.15 + 45.5)]} \text{ F} = 0.31 \mu\text{F (use 1 } \mu\text{F)}$$

Output coupling capacitor:

$$\text{Using Equ. (10-25), } C_{oc} = \frac{10}{2\pi f_L \{ (r_s \parallel R_B) / \beta + r_e \} \parallel R_E + R_L}$$

or,

$$C_{oc} = \frac{1}{2\pi \times 100[\{(600 \parallel 884)/100 + 4.3\} \parallel 500 + 100]} \text{ F} \approx 15 \mu\text{F}$$

Using Equ.(10-25),

$$C_{oc} = \frac{10}{2\pi f_L(r_e \parallel R_E + R_L)}$$

or,

$$C_{oc} = \frac{1}{2\pi \times 100(4.3 \parallel 500 + 100)} \text{ F} \approx 0.71 \mu\text{F} \text{ (use } 1 \mu\text{F)}$$

Results: The designed amplifier circuit is shown in Fig.10-12.

Comments: Due to low value of R_B , current gain of this amplifier will be about 14. To design high current gain CC amplifier, we can use transistor with high h_{fe} or Darlington pair transistor.

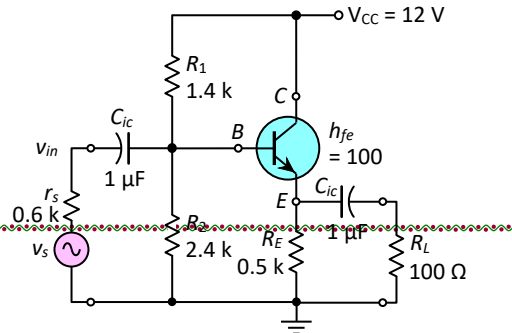


Fig.10-12: CC amplifier designed in Example 9

10.5 Single-Stage Amplifier Design with PNP Transistor

The design process of PNP transistor amplifiers is exactly same as the design process of with NPN transistor amplifiers. The design procedures discussed for CE, CB, and CC amplifiers can be used for CE, CB, and CC amplifiers with PNP transistors. Here, only the CE amplifier is discussed. Although the design considerations are same, the biasing process of PNP transistor is different. The biasing process of PNP transistor has been discussed in Chapter 4. The circuit diagram of a single-stage common emitter amplifier with PNP transistor is shown in Fig.10-13, and its design process is discussed in Example 9.

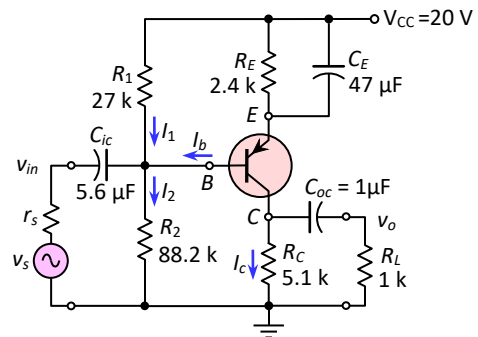


Fig.10-13: CE amplifier designed in Example 9

Example 10-7

Design a small signal CE amplifier using PNP transistor for the following specifications:

$V_{CC} = 20 \text{ V}$, $r_s = 600 \Omega$, $R_L = 3.3 \text{ k}$, lower cut-off frequency, $f_1 = 200 \text{ Hz}$, transistor $h_{fe} = 50$, $h_{ie} = 1 \text{ k}$.

Solution:

Let, $V_E = V_{CC}/4 = 5 \text{ V}$, $V_{CE} = 5 \text{ V}$ and $I_C = 2 \text{ mA}$

$$\therefore R_E = \frac{5 \text{ V}}{2 \text{ mA}} = 2.5 \text{ k} \quad \text{use } 2.4 \text{ k standard, and } V_E = 4.8 \text{ V}$$

$$V_{RC} = 20 \text{ V} - 4.8 \text{ V} - 5 \text{ V} = 10.2 \text{ V}$$

$$\therefore R_C = \frac{10.2 \text{ V}}{2 \text{ mA}} = 5.1 \text{ k} \quad 5.1 \text{ k is a standard value.}$$

Let, $I_2 = 2 \text{ mA}/10 = 0.2 \text{ mA}$. The voltage across R_1 should be, $V_{R1} = 4.8 \text{ V} + 0.7 \text{ V} = 5.5 \text{ V}$

$$\therefore R_1 = \frac{5.5 \text{ V}}{0.2 \text{ mA}} = 27.5 \text{ k} \quad \text{use standard } 27 \text{ k, and } I_2 = 0.204 \text{ mA}$$

Base current of the transistor will $I_B = 2 \text{ mA}/h_{fe} = 2 \text{ mA}/50 = 0.04 \text{ mA}$

Therefore, current through R_2 will be $I_2 = 0.204 \text{ mA} - 0.04 \text{ mA} = 0.164 \text{ mA}$

$$\therefore R_2 = \frac{V_{CC} - V_B}{I_1} = \frac{20 \text{ V} - 5.5 \text{ V}}{0.164 \text{ mA}} = 88.4 \text{ k} \quad \text{use standard } 82 \text{ k} + 6.2 \text{ k}$$

Calculation of input coupling capacitor:

$$\text{Using Equ. (10-8),} \quad C_{ic} = \frac{10}{2\pi f_1(r_s + h_{ie})} = \frac{10}{2\pi \times 200(0.6 + 1) \times 10^3}$$

$$\therefore C_{ic} = 5 \text{ } \mu\text{F, use standard } 5.6 \text{ } \mu\text{F}$$

Output Coupling capacitor:

$$\text{Using Equ. (10-10),} \quad C_{oc} = \frac{10}{2\pi f_L(R_C + R_L)} = \frac{10}{2\pi \times 200(5.1 + 3.3) \times 10^3}$$

$$\therefore C_{oc} = 0.95 \text{ } \mu\text{F, use standard } 1 \text{ } \mu\text{F}$$

Bypass Capacitor:

$$\text{Using Equ. (10-7),} \quad C_E = \frac{1 + h_{fe}}{2\pi f_L h_{ie}} = \frac{1 + 50}{2\pi \times 200 \text{ Hz} \times 10^3} = 40.6 \text{ } \mu\text{F}$$

$$\text{or,} \quad C_E = 40.6 \text{ } \mu\text{F use } 47 \text{ } \mu\text{F}$$

Result: The design values are shown in Fig.10-13.

Comments: Here, the Example 9- has been repeated for PNP transistor. All results are same except R_1 , and R_2 . We could have used the same R_1 , and R_2 by interchanging their position (R_1 up and R_2 down).

10.6 Single Stage Common Source Amplifier

Like the BJT CE amplifiers, FET common-source amplifiers are mostly used. In Chapter 8, we have found that CS amplifier has high voltage gain (though less than BJT amplifiers), high current gain, high input impedance, and high output impedance. Due to its high input impedance, CS amplifiers are generally used at the first stage of an electronic system. Like the CE amplifier, CS amplifier also produces 180° phase difference between the input and output signals.

The operation of this amplifier is shown in Fig.10-14. Here, the changes of voltage and current levels at different points of the amplifier are shown. The input signal (v_s) is considered as a sinusoidal AC signal with very small peak-to-peak value. When this signal passes through the input-coupling capacitor (C_{ic}), it superimposes with the gate-voltage V_G . So the gate-voltage changes in-phase with the input signal. When the input signal increases, the drain-current also increases ($\because I_d = g_m v_{gs}$). When drain-current increases, the voltage-drop across the drain-resistor (R_D) increases. Therefore, the voltage level at the drain terminal (that was initially V_{DQ}) decreases. As the output is taken from the drain, the output voltage decreases. The reverse process occurs when the input signal decreases. As a small change in input voltage produces a large change in drain current (and the value of R_D is large), an amplified voltage is produced at the collector. The drain-voltage swings in the same frequency and shape of the input signal but 180° out of phase. The Q-point of the amplifier oscillates with

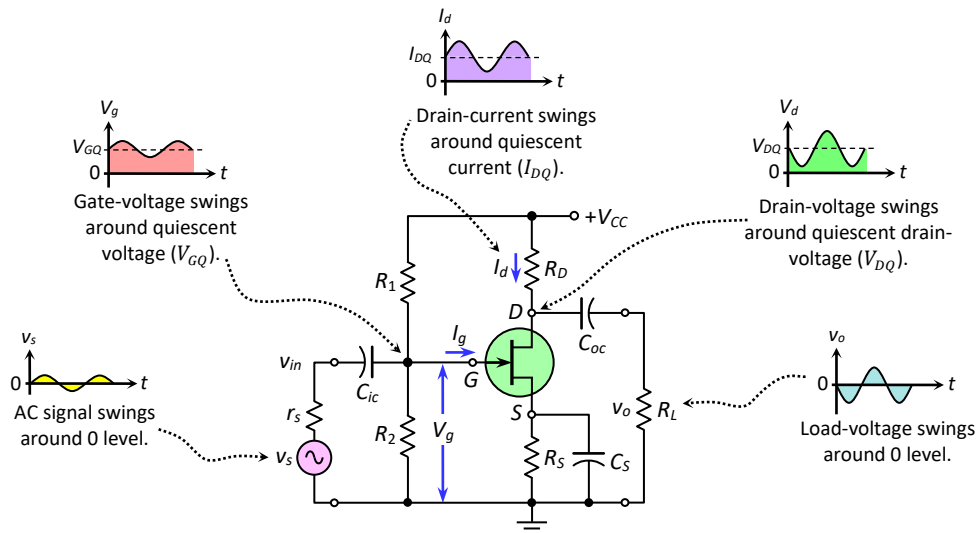


Fig.10-14: Waveforms of current and voltage swings at different points of CS amplifier

the input signal as described in Chapter 7. The amplified signal at the drain terminal has a DC voltage level as shown in Fig.10-14. When this signal passes through the output-coupling capacitor (C_{oc}), the capacitor blocks the DC voltage and the load resistor gets the pure amplified AC signal.

10.6.1 Single Stage Common Source Amplifier Design

For designing a CS amplifier generally, the following specifications are given:

The supply voltage, minimum voltage gain, frequency range, signal source impedance, and the load impedance.

Unless otherwise specified, here also voltage divider bias circuit will be used to design FET amplifiers.

Selection of I_D , R_D and R_S :

In Chapter , we have derived the equation of voltage gain of CS amplifier, which is,

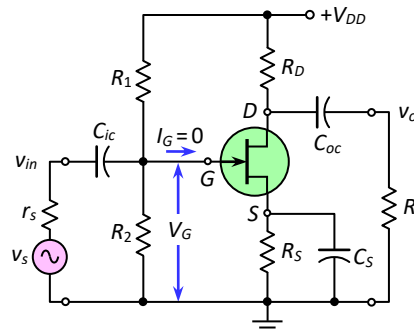


Fig. 9.15: FET Common source amplifier circuit

$$A_v = -g_m(R_D \parallel R_L) \quad (10-26)$$

From this expression we find that, larger value of R_D gives higher gain. But too large value of R_D will reduce the value of V_{DS} and the FET may work in voltage-control-resistance region. Generally, R_D should be smaller than R_L to make the gain independent of R_L . Higher values of R_D results in lower I_D . But, I_D should be high because g_m is high at higher values of I_D . So, we have to make a tradeoff between I_D and R_D . I_D can be determined from the transfer characteristic curves as described in biasing sections. The value of I_D has to be selected in such a way that the most linear portion of the transfer characteristic curve is used.

When works as an amplifier, FET must work in the pinch-off region. To ensure the operation of the FET in this region, the minimum value of V_{DS} should be $V_{P(max)} + 1\text{ V} - V_{GS}$. The rest of V_{DD} will be divided between R_D and R_S . Remember that for this value of V_{DS} , $\pm 1\text{ V}$ swing of AC signal is possible at the drain. If we need AC voltage swing more than $\pm 1\text{ V}$, then more voltage has to be allocated for V_{DS} .

Once, I_D and V_{RS} is selected, V_{GS} and R_S can be calculated from transfer characteristic curve. Then, the values of R_1 and R_2 can be calculated as described in biasing section, considering $R_1 = 1\text{ M}\Omega$.

Coupling Capacitor:

As discussed in AC analysis of FET, the voltage gain of CS amplifier with unbypassed capacitor is [Equ. (8-28)],

$$A_v = -\frac{g_m(R_D \parallel R_L)}{1 + g_m R_S} \quad (10-27)$$

Now considering the reactance of the coupling capacitor in parallel with R_S ,

$$A_v = -\frac{g_m(R_D \parallel R_L)}{1 + g_m R_S \parallel X_{CS}}$$

Normally, $R_S \gg X_{CS}$.

$$\therefore A_v = -\frac{g_m(R_D \parallel R_L)}{1 + g_m X_{CS}}$$

$$\therefore \text{magnitude of gain is, } |A_v| = \frac{g_m(R_D \parallel R_L)}{\sqrt{1^2 + (g_m X_{CS})^2}} \quad (10-28)$$

At mid-frequency range, when $X_{CS} = 0$, the above equation gives the flat voltage gain as,

$$|A_{v(mid)}| = \frac{g_m(R_D \parallel R_L)}{\sqrt{1^2}} = g_m(R_D \parallel R_L) \quad (10-29)$$

As the frequency of the signal decreases, the reactance of the capacitor increases and the gain of the amplifier decreased from the mid-band gain. At some frequency, when, $g_m X_{CS} = 1$, from Equ.(10-28) we get,

$$|A_v| = \frac{g_m(R_D \parallel R_L)}{\sqrt{1^2 + 1^2}} = \frac{g_m(R_D \parallel R_L)}{\sqrt{2}} = \frac{A_{v(mid)}}{\sqrt{2}}$$

As the gain falls from the mid-frequency gain by a factor of $1/\sqrt{2}$, this frequency is called the lower cut-off frequency (f_L).

Therefore, the condition for lower cut-off frequency is,

$$g_m X_{CS} = 1 \quad \text{or,} \quad X_{CS} = \frac{1}{g_m} \text{ at } f_L$$

$$\text{or,} \quad \frac{1}{2\pi f_L C_S} = \frac{1}{g_m}$$

$$\therefore \quad \boxed{C_S = \frac{g_m}{2\pi f_L}} \quad (10-30)$$

Input Coupling Capacitor:

Like the BJT CE amplifier, the bypass capacitor of CS amplifier will have the largest value. So, the lower cut-off frequency will be determined by this bypass capacitor. The reactance of the input coupling capacitor and the output coupling capacitor should be so small that they do not further reduce the gain of the amplifier. A reasonable result will be obtained, if the reactance of the coupling capacitor is made ten times lower than the series resistance connected with the capacitor. Therefore, the input coupling capacitor can be calculated as,

$$X_{Cic} = \frac{r_s + Z_i}{10} \text{ at } f_L$$

$$\text{or,} \quad \frac{1}{2\pi f_L C_{ic}} = \frac{r_s + Z_i}{10}$$

$$\text{or,} \quad C_{ic} = \frac{10}{2\pi f_L (r_s + Z_i)}$$

For CS amplifier, $Z_i \approx R_G = R_1 \parallel R_2$.

$$\therefore \quad \boxed{C_{ic} = \frac{10}{2\pi f_L (r_s + Z_i)}} \quad (10-31)$$

Output Coupling Capacitor:

The output coupling capacitor can be calculated as,

$$X_{Coc} = \frac{Z_o + R_L}{10} \text{ at } f_L$$

$$\text{or,} \quad \frac{1}{2\pi f_L C_{oc}} = \frac{Z_o + R_L}{10}$$

$$\text{or,} \quad C_{oc} = \frac{10}{2\pi f_L (Z_o + R_L)}$$

For CS amplifier, $Z_o = R_D$.

$$\therefore \quad \boxed{C_{oc} = \frac{10}{2\pi f_L (R_D + R_L)}} \quad (10-32)$$

Another approach to calculate the coupling capacitor is to making $X_{C_{ic}} = Z_i$ at two octaves below f_L , that is

$$X_{C_{ic}} = Z_i \text{ at } f_L/16$$

or,

$$\frac{16}{2\pi f_L C_{ic}} = Z_i$$

∴

$$C_{ic} = \frac{16}{2\pi f_L Z_i} \quad (10-33)$$

Similarly, the output coupling capacitor can be calculated making $X_{C_{oc}} = R_L$ at two octaves below f_L , that is

$$C_{oc} = \frac{16}{2\pi f_L R_L} \quad (10-34)$$

Example 10-8

Suppose a JFET has $V_{P(\max)} = -5 \text{ V}$, and $V_{P(\min)} = -2 \text{ V}$, $I_{DSS(\max)} = 18 \text{ mA}$ and $I_{DSS(\min)} = 8 \text{ mA}$. Design a CS amplifier considering $V_{DD} = 25 \text{ V}$, lower cut-off frequency = 200 Hz, $R_L = 5.1 \text{ k}$ and $r_s = 600 \Omega$.

Solution:

As shown in the transfer characteristic curve, the maximum drain current, $I_{D(\max)}$, can be 5 mA to 7 mA. Let, $I_{D(\max)} = 5 \text{ mA}$.

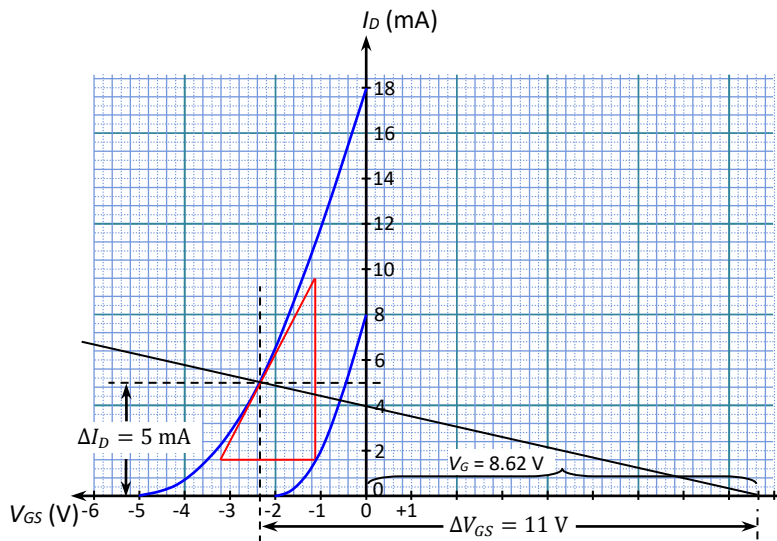


Fig.10-16: Transfer characteristics and bias current for Example 9.8

Rearranging Shockley's equation, value of V_{GS} can be calculated as,

$$V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}}) = -5(1 - \sqrt{5/18}) = -2.4 \text{ V}$$

Minimum drain-source voltage will be,

$$V_{DS(\min)} = V_{P(\max)} + 1 \text{ V} - V_{GS} = 5 + 1 - 2.4 \approx 3.6 \text{ V}$$

Let, rest of the voltage equally divide between R_D , and R_S .

$$V_{RS} = V_{RD} = \frac{1}{2}(V_{DD} - V_{DS(\min)}) = \frac{1}{2}(25 \text{ V} - 3.6 \text{ V}) = 10.7 \text{ V}$$

$$\therefore R_S = R_D = \frac{10.7 \text{ V}}{5 \text{ mA}} = 2.14 \text{ k}, \text{ use standard } 2.2 \text{ k}$$

Determination of bias line:

For a fixed value of V_G , V_{GS} will change with I_D as,

$$\Delta V_{GS} = \Delta I_D R_S$$

$$\text{Let, } \Delta I_D = 5 \text{ mA} \quad \therefore \Delta V_{GS} = 5 \text{ mA} \times 2.2 \text{ k} = 11 \text{ V}$$

Using these changes the bias line is drawn in Fig.10-16. From the graph $V_G = 8.62 \text{ V}$

$$\text{Let, } R_1 = 1 \text{ M}\Omega, \quad \therefore I_2 = \frac{8.62 \text{ V}}{1 \text{ M}\Omega}$$

$$\therefore R_2 = \frac{V_{DD} - V_G}{I_2} = \frac{25 \text{ V} - 8.62 \text{ V}}{8.62 \text{ V}} \times 1 \text{ M}\Omega \approx 2 \text{ M}\Omega$$

The value of g_m at the bias point can be calculated by drawing tangent at this point as shown in the figure.

$$g_m = \frac{8 \text{ mA}}{2.1 \text{ V}} = 3.8 \text{ mS}$$

Alternately, this value can also be calculated using Equ.(6-6) as,

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \left|\frac{V_{GS}}{V_P}\right|\right) = \frac{2 \times 18 \text{ mA}}{5 \text{ V}} \left(1 - \frac{2.4 \text{ V}}{5 \text{ V}}\right) \approx 3.74 \text{ mS}$$

Bypass capacitor:

Using Equ.(10-30),

$$C_S = \frac{g_m}{2\pi f_L} = \frac{3.8 \times 10^{-3}}{2\pi \times 200} = 3 \text{ }\mu\text{F}, \text{ use standart } 3.3 \text{ }\mu\text{F}$$

Input impedance $Z_i = R_G = R_1 \parallel R_2 = 1 \text{ M} \parallel 2 \text{ M} = 667 \text{ k}$

Using Equ.(10-31),

$$C_{ic} = \frac{10}{2\pi f_L(r_s + Z_i)} = \frac{10}{2\pi \times 200(0.6 \text{ k} + 667 \text{ k})} \approx 12 \text{ nF}$$

Output coupling capacitor:

Using Equ.(10-31),

$$C_{oc} = \frac{10}{2\pi \times 200(2.2 \text{ k} + 5.1 \text{ k})}$$

or, $C_{oc} = 1.1 \text{ } \mu\text{F}$ (use $1.2 \text{ } \mu\text{F}$)

Result: The design amplifier is shown in Fig.10-17.

Comments: Here, the value of gate voltage (V_G) has been calculated graphically, this value can also be calculated analytically.

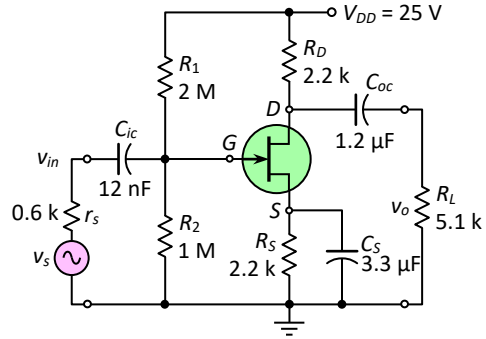


Fig. 9.17: CS amplifier designed in Example 9.8

10.7 Single Stage Common-Drain Amplifier Design

Like common-source amplifier, for designing a common-drain amplifier generally, the following specifications are given:

The supply voltage, minimum voltage gain, frequency range, signal source impedance, the load impedance etc.

Although self-bias can be used for CD amplifier, voltage divider bias is most commonly used. Fig.10-18 shows a common-drain amplifier with voltage divider bias. The design technique is described below.

Selection of I_D , R_D and R_S :

We know that the voltage gain of a CD amplifier is unity. But the actual voltage gain of this amplifier, as derived in Chapter 8 [Equ.8-37], is

$$A_v = \frac{g_m(R_S \parallel R_L)}{1 + g_m(R_S \parallel R_L)} \quad (10-35)$$

To keep the voltage gain of this amplifier close to unity, the value of $g_m(R_S \parallel R_L)$ should be much greater than 1. We know that the value of g_m of a small signal FET is few mS. Therefore, the value of $R_S \parallel R_L$ should be few kilo-ohms or higher. Otherwise the voltage gain will be quite less than unity.

As CD amplifier is used buffer, the current gain of this amplifier should be high. The expression of current gain of this amplifier is,

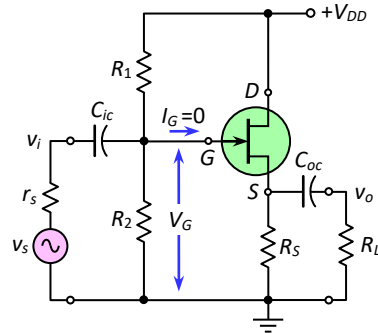


Fig.10-18: CD amplifier with voltage divider bias

$$A_i = \frac{g_m R_S R_G}{(R_L + R_S)[1 + g_m(R_S \parallel R_L)]} \quad (10-36)$$

From this expression it can be proved that, the larger value of R_S , smaller values of R_L , and $R_S \parallel R_L$ will give better current gain. By increasing R_G we can increase the current gain as well. , in most FET biasing circuits, a very large value of R_G is used.

As smaller value of $R_S \parallel R_L$ increase current gain and decrease voltage gain, we have to make a trade-off, or design the current according to our requirements.

Like the CS amplifier, in CD amplifier, I_D should be high because g_m is high at higher values of I_D . The value of I_D has to be selected in such a way that the most linear portion of the transfer characteristic curve is used and AC current swing has sufficient space.

To ensure the operation of the FET in the pinch-off region, the minimum value of V_{DS} should be $V_{P(max)} + 1\text{ V} - V_{GS}$. The rest of V_{DD} will go across R_S . However, to get maximum voltage swing, we can also set $V_{DS} = V_{RS} = V_{DD}/2$.

Once, I_D and V_{RS} is selected, R_S can be calculated using Ohm's law. Using the value of I_D on the transfer characteristic curve V_{GS} can be calculated. The value of gate voltage will be, $|V_G| = |V_S| + |V_{GS}|$. Using V_G , the values of R_1 and R_2 can be calculated as described in biasing section, considering $R_1 = 1\text{ M}\Omega$.

Output Coupling Capacitor:

In common-drain amplifier there are only two coupling capacitors, and no bypass capacitor. Among the two capacitors, the output coupling capacitor will have the larger value. Therefore, the lower cut-off frequency will be determined by the output coupling capacitor. So, we have to calculate its value by equating its reactance to the series resistance. From the Fig.10-2 we find that the resistance connected in series with C_{oc} are the output impedance of the amplifier (Z_o), and the load resistance (R_L). Therefore,

$$X_{C_{oc}} = Z_o + R_L \text{ at } f_L$$

$$\text{or,} \quad \frac{1}{2\pi f_L C_{oc}} = Z_o + R_L$$

For common-collector amplifier the output impedance is,

$$Z_o = \frac{1}{g_m} \parallel R_S = R_S / (1 + g_m R_S)$$

$$\therefore \quad C_{oc} = \frac{1}{2\pi f_L [R_S / (1 + g_m R_S) + R_L]} \quad (10-37)$$

Input Coupling Capacitor:

The cut-off frequency of CD amplifier is determined by the output coupling capacitor (C_{oc}). So the reactance of the input coupling capacitor should be very small, otherwise the output voltage will be further attenuated. Therefore, its value will be calculated considering its reactance 10 times smaller than the total series resistance in its path. As shown in Fig.10-2, the resistance connected in series with C_{ic} are the source resistance (r_s) and the input impedance of the amplifier (Z_i). Therefore,

$$X_{C_{ic}} = \frac{r_s + Z_i}{10} \text{ at } f_L$$

The input impedance of a CD amplifier is, $Z_{in} \approx R_G$

$$\therefore \frac{1}{2\pi f_L C_{ic}} = \frac{r_s + R_G}{10}$$

$$\therefore \boxed{C_{ic} = \frac{10}{2\pi f_L (r_s + R_G)}} \quad (10-38)$$

Example 10-9

Suppose a JFET has $V_{P(max)} = -4$ V and $V_{P(min)} = -2$ V, and $I_{DSS(max)} = 16$ mA and $I_{DSS(min)} = 8$ mA. Design a CD amplifier considering $V_{DD} = 20$ V, lower cut-off frequency = 100 Hz, $R_L = 4.3$ k and $r_s = 600$ Ω .

Solution:

The maximum and minimum transfer characteristic curves are drawn in Fig.10-19. From this figure a suitable maximum drain current can be $I_{D(max)} = 6$ mA.

$$\text{Let,} \quad V_S = V_{DS} = \frac{1}{2} V_{DD} = \frac{1}{2} 20 \text{ V} = 10 \text{ V}$$

$$\therefore R_S = \frac{V_S}{I_S} = \frac{V_S}{I_D} = \frac{10 \text{ V}}{6 \text{ mA}} \approx 1.67 \text{ k use } 1.6 \text{ k}$$

$$\therefore V_S = R_S I_D = 1.6 \text{ k} \times 6 \text{ mA} = 9.6 \text{ V}$$

$$\text{and,} \quad V_{DS} = V_{DD} - V_S = 20 \text{ V} - 9.6 \text{ V} = 10.4 \text{ V}$$

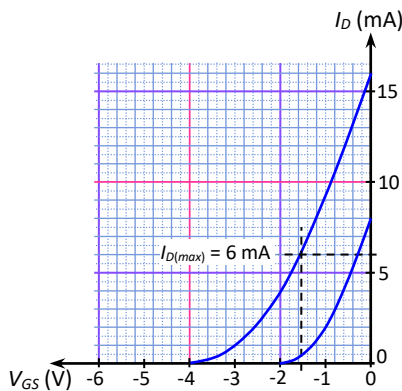


Fig. 9.19: Transfer characteristics and bias current for Example 9.9

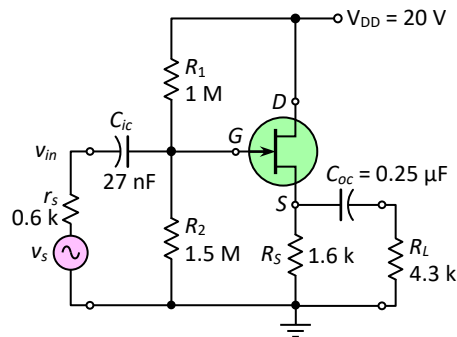


Fig. 9.20: CD amplifier designed in Example 9.9

For $I_{D(max)} = 6$ mA, the required gate-source voltage can be calculated from Shockley's equation as,

$$V_{GS} = V_P \left(1 - \sqrt{I_D / I_{DSS}} \right)$$

or,
$$V_{GS} = -4 \text{ V} (1 - \sqrt{6 \text{ mA} / 16 \text{ mA}}) \approx -1.55 \text{ V}$$

∴
$$V_G = V_S - |V_{GS}| = 9.6 \text{ V} - 1.55 \text{ V} = 8.05 \text{ V}$$

Calculation of bias resistors:

Let,
$$R_1 = 1 \text{ M}$$

∴
$$R_2 = \frac{R_1 \times V_{R2}}{V_{R1}} = \frac{1 \text{ M} \times 11.96 \text{ V}}{8.05 \text{ V}} \approx 1.5 \text{ M}$$

Input coupling capacitor:

Using Equ.(10-38),
$$C_{ic} = \frac{10}{2\pi f_L (r_s + R_G)} = \frac{10}{2\pi f_L (r_s + R_1 \parallel R_2)}$$

or,
$$C_{ic} = \frac{10}{2\pi 100 \text{ Hz} (600 + 1 \text{ M} \parallel 1.5 \text{ M})} \approx 27 \text{ nF}$$

Output coupling capacitor:

At $I_{D(max)} = 6 \text{ mA}$, the value of g_m can be calculated using Equ.6-11,

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} = \frac{2\sqrt{16 \text{ mA} \times 6 \text{ mA}}}{4 \text{ V}} \approx 4.9 \text{ mS}$$

Using Equ.(10-37),
$$C_{oc} = \frac{1}{2\pi f_L [R_S / (1 + g_m R_S) + R_L]}$$

or,
$$C_{oc} = \frac{10}{2\pi 100 \text{ Hz} [600 / (1 + 4.9 \text{ mS} \times 1.6 \text{ k})]} \approx 0.25 \text{ nF}$$

Result: The design amplifier in this example is shown in Fig.10-20.

Comments: As the output impedance is very large the output coupling capacitor has small value.



10.8 Single Stage Common-Gate Amplifier Design

Like the other two FET amplifiers, the design of a common-gate amplifier may start with some specifications. These are:

The supply voltage, minimum voltage gain, frequency range, signal source impedance, the load impedance etc.

In Chapter 7, we have seen that both self-bias and voltage divider bias can be used for common-gate amplifier. As the voltage divider bias gives better stability of the operating point, it is most commonly used. A common-gate amplifier with voltage divider bias is shown in Fig.10-21.

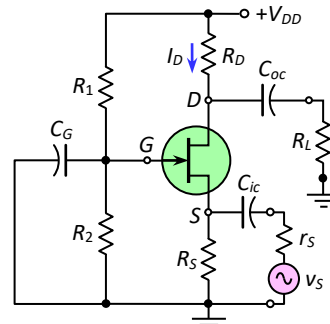


Fig.10-21: CG amplifier with voltage divider bias

Design technique of this amplifier is described below.

Selection of I_D , R_D and R_S :

In Chapter 7, we have derived the voltage gain [Equ.8-37] of a CG amplifier. It is,

$$A_v = g_m R_D \parallel R_L \quad (10-39)$$

To achieve high voltage gain, the value of g_m and $(R_D \parallel R_L)$ should be high. We know that the value of g_m is higher at higher values of I_D . Similarly, R_D should be large to get high voltage gain. If the load resistance, R_L is fixed, increasing R_D beyond R_L will not increase the voltage gain so much. Therefore, we can select the value of R_D between R_L to $5R_L$. However, if we can set $R_D < R_L$, without sacrificing the voltage gain (eg., if R_L is too large), then the gain of the amplifier will be comparatively less effected by R_L . Moreover, smaller values of R_D than R_L will give better current gain. The actual selection of R_D will depend on the values of I_D and the voltage allocated across R_D .

Like the CS, and CD amplifier, in CG amplifier, I_D should have as high value as possible, because g_m is higher at higher values of I_D . The value of I_D has to be selected in such a way that the most linear portion of the transfer characteristic curve is used and the AC voltage swing has sufficient space. As described in Chapter 7, in designing the FET circuit, we have to consider the maximum transfer characteristic curve of the FET. Once I_D is selected, the required value of V_{GS} can be calculated from the transfer characteristic curve or using Equ.

To ensure the operation of the FET in the pinch-off region, the minimum value of V_{DS} should be $|V_{P(max)}| + 1 \text{ V} - |V_{GS}|$. We can divide the rest of the voltage between R_D and R_S , either equally or can give a little more voltage to R_D than R_S to increase the gain. Now, the values of R_D , and R_S can be calculated using Ohm's law.

The value of gate voltage will be, $|V_G| = |V_S| - |V_{GS}|$. Using V_G , the values of R_1 and R_2 can be calculated. Generally we set $R_1 = 1 \text{ M}\Omega$. Therefore,

$$R_2 = \frac{R_1(V_{DD} - V_G)}{V_G}$$

Bypass Capacitor:

As shown if Fig, CG amplifier with voltage divider bias needs a bypass capacitor at the gate terminal. It shorts the gate to ground for AC signals. The series resistance seen by C_G is the bias resistance ($R_G = R_1 \parallel R_2$) in parallel with the FET internal resistance at gate terminal (R_{gs}), that is $R_G \parallel R_{gs}$. Therefore, the value of C_G can be calculated as,

$$X_{CG} = \frac{R_G \parallel R_{gs}}{10}$$

$$\text{or,} \quad \frac{1}{2\pi f_L C_G} = \frac{R_G \parallel R_{gs}}{10}$$

$$\text{or,} \quad C_G = \frac{10}{2\pi f_L R_G \parallel R_{gs}}$$

$$\because R_{gs} \gg R_G \quad C_G = \frac{10}{2\pi f_L R_G} \quad (10-40)$$

The value of C_G will be so small that we can neglect this bypass capacitor.

Input Coupling Capacitor:

In common-gate amplifier, the input coupling capacitor will have the largest value. Therefore, the lower cut-off frequency will be determined by this input coupling capacitor. So, we have to calculate its value by considering its reactance equal to the series resistance connected in its path. From Fig.10-2 we find that the resistance connected in series with C_{ic} are the source resistance r_s and the input impedance of the amplifier (Z_i). Therefore,

$$X_{C_{ic}} = Z_i + r_s \text{ at } f_L$$

$$\text{or, } \frac{1}{2\pi f_L C_{ic}} = Z_i + r_s$$

For common-gate amplifier the input impedance is,

$$Z_i = \frac{1}{g_m} \parallel R_S = R_S / (1 + g_m R_S)$$

$$\therefore \boxed{C_{ic} = \frac{1}{2\pi f_L [R_S / (1 + g_m R_S) + r_s]}} \quad (10-41)$$

Output Coupling Capacitor:

The cut-off frequency of CG amplifier is determined by the input coupling capacitor (C_{ic}). So the reactance of the output coupling capacitor should be very small, otherwise the voltage gain will be further attenuated. So, the value of C_{oc} will be calculated considering its reactance 10 times smaller than the total series resistance in its path. From Fig.10-2, the resistance connected in series with C_{oc} are the output impedance of the amplifier (Z_o), and the load resistance (R_L). Therefore,

$$X_{C_{oc}} = \frac{Z_o + R_L}{10} \text{ at } f_L$$

The output impedance of a CG amplifier is, $Z_o \approx R_D$

$$\therefore \frac{1}{2\pi f_L C_{oc}} = \frac{R_D + R_L}{10}$$

$$\text{or, } \boxed{C_{oc} = \frac{10}{2\pi f_L (R_D + R_L)}} \quad (10-42)$$

Example 10-10

JFET 2N5951 has $V_{P(max)} = -5 \text{ V}$ and $V_{P(min)} = -2 \text{ V}$, and $I_{DSS(max)} = 13 \text{ mA}$ and $I_{DSS(min)} = 7 \text{ mA}$. Design a CG amplifier considering $V_{DD} = 30 \text{ V}$ and lower cut-off frequency = 100 Hz, $R_L = 5.1 \text{ k}$ and $r_s = 600 \Omega$. The minimum gain of the amplifier should be 6.

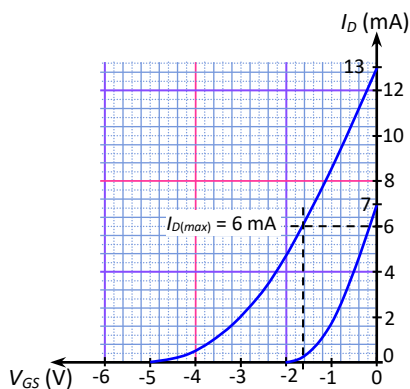


Fig.10-22: Transfer characteristics and bias current for Example 9.10

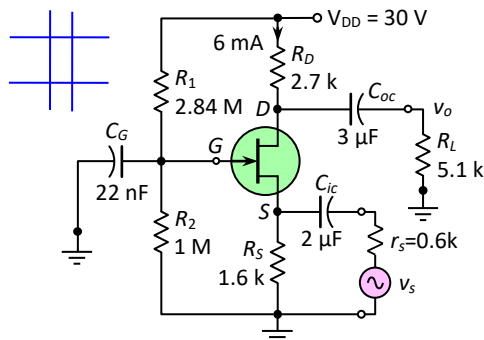


Fig.10-23: CG amplifier designed in Example 9.10

Solution:

The maximum and minimum transfer characteristic curves are drawn in Fig.10-22. From this figure a suitable maximum drain current can be $I_{D(max)} = 6 \text{ mA}$.

For $I_{D(max)} = 6 \text{ mA}$, the required gate-source voltage can be calculated from Shockley's equation as,

$$V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}})$$

$$\text{or, } V_{GS} = -5 \text{ V}(1 - \sqrt{6 \text{ mA} / 13 \text{ mA}}) \approx -1.6 \text{ V}$$

and

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} = \frac{2\sqrt{13 \text{ mA} \times 6 \text{ mA}}}{5 \text{ V}} \approx 3.5 \text{ mS}$$

Required value of R_D to achieve voltage gain of 6 can be calculated as,

$$A_v = 6 = g_m R_D \parallel R_L = 3.5 \text{ mS} \times R_D \parallel R_L$$

or,

$$R_D \parallel R_L = 1.7 \text{ k}$$

or,

$$R_D \parallel 5.1 \text{ k} = 1.7 \text{ k}$$

∴

$$R_D = 2.55 \text{ k, use standard } 2.7 \text{ k}$$

The voltage drop across R_D will be,

$$V_{RD} = I_D R_D = 6 \text{ mA} \times 2.7 \text{ k} = 16.2 \text{ V}$$

The minimum drain-source voltage will be,

$$V_{DS(min)} = |V_{P(max)}| + 1 \text{ V} - |V_{GS}| = (5 + 1 - 1.6) \text{ V} = 4.4 \text{ V}$$

The remaining voltage will go to source. So,

$$V_S = V_{DD} - V_{DS(min)} - V_{RD} = (30 - 4.4 - 16.2) \text{ V} = 9.4 \text{ V}$$

∴

$$R_S = \frac{V_S}{I_D} = \frac{9.4 \text{ V}}{6 \text{ mA}} \approx 1.6 \text{ k}$$

Value of gate voltage will be,

$$V_G = |V_S| - |V_{GS}| = 9.4 \text{ V} - 1.6 \text{ V} = 7.8 \text{ V}$$

Let,

$$R_1 = 1 \text{ M}$$

$$\therefore R_2 = \frac{R_1(V_{DD} - V_G)}{V_G} = \frac{1 \text{ M}(30 \text{ V} - 7.8 \text{ V})}{7.8 \text{ V}} \approx 2.84 \text{ M}$$

To make this value we can use three resistors (2.2 M, 620 k, and 20 k) in series.

Input coupling capacitor:

At $I_{D(max)} = 6 \text{ mA}$, the value of g_m can be calculated using Equ.6-11,

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} = \frac{2\sqrt{13 \text{ mA} \times 6 \text{ mA}}}{5 \text{ V}} \approx 3.5 \text{ mS}$$

Using Equ.(10-41),

$$C_{ic} = \frac{1}{2\pi f_L [R_S / (1 + g_m R_S) + r_s]}$$

$$\text{or, } C_{ic} = \frac{1}{2\pi 100 \text{ Hz} [1.3 \text{ k} / (1 + 3.5 \text{ mS} \times 1.6 \text{ k}) + 600 \Omega]} \approx 2 \mu\text{F}$$

Output coupling capacitor:

$$\text{Using Equ.(10-42), } C_{oc} = \frac{10}{2\pi f_L (R_D + R_L)} = \frac{10}{2\pi 100 \text{ Hz} (2.7 \text{ k} + 5.1 \text{ k})} \approx 3 \mu\text{F}$$

Bypass capacitor:

$$\text{Using Equ.(10-40), } C_G = \frac{10}{2\pi f_L R_G} = \frac{10}{2\pi 100 \text{ Hz} \times (2.84 \text{ M} \parallel 1 \text{ M})} \approx 22 \text{ nF}$$

Results: The designed common-gate amplifier is shown in Fig.10-23.

Comment: The gate bypass capacitor is very small. So if this capacitor is not used in the circuit the gain will not be hampered.

10.9 Design Process of DMOSFET Amplifiers

In Chapter 7, the biasing techniques of DMOSFETs have been discussed. In Chapter 8, we have covered the AC analysis of DMOSFET amplifiers. It is found that the expressions for voltage gain, input impedance, output impedance, and current gain are exactly similar to those of JFET amplifiers. Therefore, DMOSFET amplifiers can be designed using the same processes discussed for JFET amplifiers.

Example 10-11

BSS139 is an N-channel DMOSFET with $V_P = -2.1 \text{ V}$, $I_{DSS} = 30 \text{ mA}$, and $R_{GS} = 10^9 \Omega$. Design a CG amplifier considering $V_{DD} = 30 \text{ V}$, $f_L = 100 \text{ Hz}$, $R_L = 1 \text{ k}$ and $r_s = 600 \Omega$. The amplifier must be capable to produce an AC voltage swing of $\pm 2 \text{ V}$ minimum.

Solution:

The transfer characteristic curve is drawn in Fig.10-24. From this figure a suitable quiescent drain current can be $I_{DQ} = 15 \text{ mA}$.

For $I_{DQ} = 15 \text{ mA}$, the required gate-source voltage can be calculated from Shockley's equation as,

$$V_{GS} = V_P(1 - \sqrt{I_D/I_{DSS}})$$

or,

$$V_{GS} = -2.1 \text{ V}(1 - \sqrt{15 \text{ mA} / 30 \text{ mA}}) \approx -0.62 \text{ V}$$

and

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} = \frac{2\sqrt{30 \text{ mA} \times 15 \text{ mA}}}{2.1 \text{ V}} \approx 20.2 \text{ mS}$$

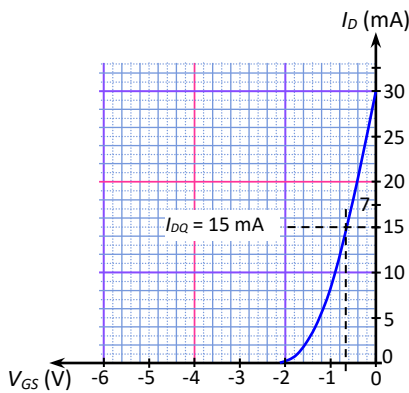


Fig.10-24: Transfer characteristics and bias current for Example 9.11

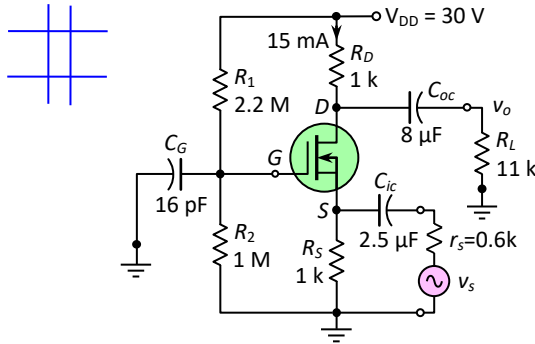


Fig.10-25: CG amplifier designed in Example 9.11

The minimum value of V_{DS} is,

$$V_{DS(min)} = |V_P| - |V_{GS}| + \text{peak swing}$$

or,

$$V_{DS(min)} = (2.1 - 0.62 + 2) \text{ V} = 3.48$$

Let,

$$V_{DS} = 5 \text{ V}, V_S = 10 \text{ V}, \text{ and } V_{RD} = 15 \text{ V}$$

∴

$$R_D = \frac{V_{RD}}{I_D} = \frac{15 \text{ V}}{15 \text{ mA}} = 1 \text{ k}$$

∴

$$R_S = \frac{V_S}{I_D} = \frac{10 \text{ V}}{15 \text{ mA}} = 667 \Omega, \text{ use } 680 \Omega$$

Actual value of V_S ,

$$V_S = I_D R_S = 15 \text{ mA} \times 680 \Omega = 10.2 \text{ V}$$

Value of gate voltage will be,

$$V_G = |V_S| - |V_{GS}| = 10.2 \text{ V} - 0.62 \text{ V} = 9.58 \text{ V}$$

Let,

$$R_1 = 1 \text{ M}$$

∴

$$R_2 = \frac{R_1(V_{DD} - V_G)}{V_G} = \frac{1 \text{ M}(30 \text{ V} - 9.58 \text{ V})}{9.58 \text{ V}} \approx 2.13 \text{ M}, \text{ use } 2.2 \text{ M}$$

Input coupling capacitor:

Using Equ.(10-41)
$$C_{ic} = \frac{1}{2\pi f_L [R_S / (1 + g_m R_S) + r_s]}$$

or,
$$C_{ic} = \frac{1}{2\pi 100 \text{ Hz} [680 \Omega / (1 + 20.2 \text{ mS} \times 0.68 \text{ k}) + 600 \Omega]} \approx 2.5 \mu\text{F}$$

Output coupling capacitor:

Using Equ.(10-42),
$$C_{oc} = \frac{10}{2\pi f_L (R_D + R_L)} = \frac{10}{2\pi 100 \text{ Hz} (1 \text{ k} + 1 \text{ k})} \approx 8 \mu\text{F}$$

Bypass capacitor:

Using Equ.(10-40),
$$C_G = \frac{10}{2\pi f_L R_{GS}} = \frac{10}{2\pi 100 \text{ Hz} \times 10^9} \approx 16 \text{ pF}$$

Results: The designed common-gate amplifier is shown in Fig.10-25.

Comment: The gate bypass capacitor is very small. So if this capacitor is not used in the circuit the gain will not be hampered so much.

10.10 Design Process of EMOSFET Amplifiers

The design process of EMOSFET amplifier is almost similar to the design process of JFET amplifiers. Here, the main difference is the value of $V_{DS(\min)}$. We know that the drain-source saturation voltage of EMOSFET device is (Equ.6-11),

$$V_{DS(\text{sat})} = V_{GS} - V_T$$

Therefore, to get the required amount of AC voltage swing the value of $V_{DS(\min)}$ should be,

$$V_{DS(\min)} = V_{DS(\text{sat})} + \text{peak value of AC swing} + 1 \text{ V} \quad (10-43)$$

Another difference is, here the gate voltage should be greater than source voltage to ensure the quiescent drain current. The value of V_G can be calculated as,

$$V_G = V_S + V_{GS} \quad (10-44)$$

Alternately, the value can be determined from the transfer characteristic curve by drawing the bias line.

Example 10-12

2N7002-7-F is a small signal EMOSFET. Some of the parameters of this device are shown below. Design a CS amplifier considering $f_L = 100 \text{ Hz}$, $R_L = 1 \text{ k}$ and $r_s = 600 \Omega$. The minimum gain of the amplifier should be 50 and the amplifier should produce an AC swing of $\pm 2 \text{ V}$ at the output. Suggest the value of required V_{DD} .

Parameters	Symbol	Values	Unit
------------	--------	--------	------

		Min.	Typical	Max.	
Drain-Source breakdown voltage	V_{DSS}	60			V
Gate Threshold voltage	$V_{GS(th)}$	1	2.1	2.5	V
On-state drain current	$I_{D(on)}$	500	2700		mA
Forward trans conductance	g_{FS} (at $I_D = 200$ mA)	80	320		mS

Solution:

To draw the transfer characteristic curves we need the value of k . The value of k can be calculated using Equ.6-21.

$$g_m = 2\sqrt{I_D k}$$

or,

$$k = \frac{g_m^2}{4I_D}$$

$\therefore$

$$k_{\min} = \frac{(80 \text{ mS})^2}{4 \times 200 \text{ mA}} = 8 \text{ mA/V}^2$$

$\therefore$

$$k_{\text{typical}} = \frac{(320 \text{ mS})^2}{4 \times 200 \text{ mA}} = 128 \text{ mA/V}^2$$

Now, using these values of k and the given data of the FET, the minimum and typical transfer characteristic curves are drawn in Fig.10-26. From this figure a suitable maximum drain current can be $I_{D(max)} = 40$ mA. To get better linearity and g_m , we could have set larger value of $I_{D(max)}$ but, larger value of drain current will result in power loss.

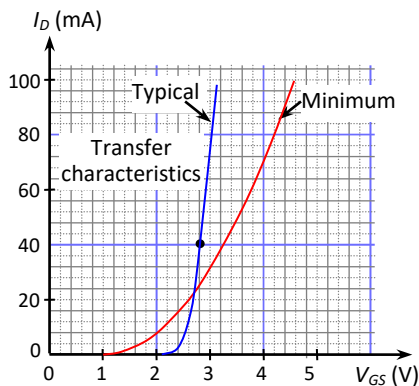


Fig. 10-26: Transfer characteristics and bias current for Example 9.12

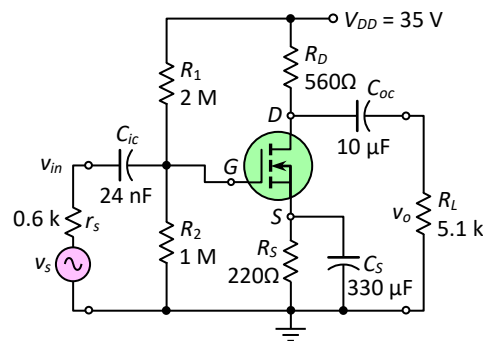


Fig. 9.27: CS amplifier designed in Example 10.12

For $I_{D(max)} = 40$ mA, the typical value of g_m will be,

$$g_m = 2\sqrt{I_D k} = 2\sqrt{40 \text{ mA} \times 128 \text{ mA/V}^2} \approx 143 \text{ mS}$$

and,

$$I_D = k(V_{GS} - V_T)^2$$

or,

$$V_{GS} = \sqrt{I_D/k} + V_T = \sqrt{40 \text{ mA}/128 \text{ mA/V}^2} + 2.1 \approx 2.66 \text{ V}$$

Required value of R_D to achieve a voltage gain of 20 can be calculated as,

$$A_v = 50 = g_m R_D \parallel R_L = 143 \text{ mS} \times R_D \parallel R_L$$

or,

$$R_D \parallel R_L = 350 \Omega$$

or,

$$R_D \parallel 1000 \Omega = 350 \Omega$$

$\therefore$

$$R_D = 539 \Omega, \text{ use } 560 \Omega$$

The voltage drop across R_D will be,

$$V_{RD} = I_D R_D = 40 \text{ mA} \times 0.56 \text{ k} = 22.4 \text{ V}$$

The drain-source saturation voltage will be,

$$V_{DS(sat)} = |V_{GS}| - |V_T| = 2.66 \text{ V} - 2.1 \text{ V} = 0.56 \text{ V}$$

To achieve $\pm 2 \text{ V}$ swing, the minimum drain-source voltage should be,

$$V_{DS(min)} = V_{DS(sat)} + \text{swing} + 1 \text{ V} = 0.56 \text{ V} + 2 \text{ V} + 1 = 3.56 \text{ V}$$

Let,

$$V_S = 9 \text{ V}$$

Therefore,

$$V_{DD} = V_S + V_{DS(min)} + V_{RD} = (9 \text{ V} + 3.56 + 22.4) \text{ V} \approx 35 \text{ V}$$

The source resistance will be,

$$\therefore R_S = \frac{V_S}{I_D} = \frac{9 \text{ V}}{40 \text{ mA}} \approx 225 \Omega, \text{ use } 220 \Omega$$

Exact value of source voltage will be,

$$\therefore V_S = 40 \text{ mA} \times 220 \Omega = 8.8 \text{ V}$$

Value of gate voltage will be

$$V_G = |V_S| + |V_{GS}| = 8.8 \text{ V} + 2.66 \text{ V} = 11.46 \text{ V}$$

Let,

$$R_1 = 1 \text{ M}$$

$\therefore$

$$R_2 = \frac{R_1(V_{DD} - V_G)}{V_G} = \frac{1 \text{ M}(35 \text{ V} - 11.46 \text{ V})}{11.46 \text{ V}} \approx 2 \text{ M}$$

Bypass capacitor:

Using Equ.(10-30),

$$C_S = \frac{g_m}{2\pi f_L} = \frac{143 \times 10^{-3}}{2\pi \times 100} = 227.6 \mu\text{F}, \text{ use standart } 330 \mu\text{F}$$

Input coupling capacitor:

Input impedance $Z_i = R_G = R_1 \parallel R_2 = 1 \text{ M} \parallel 2 \text{ M} = 667 \text{ k}$

Using Equ.(10-31),

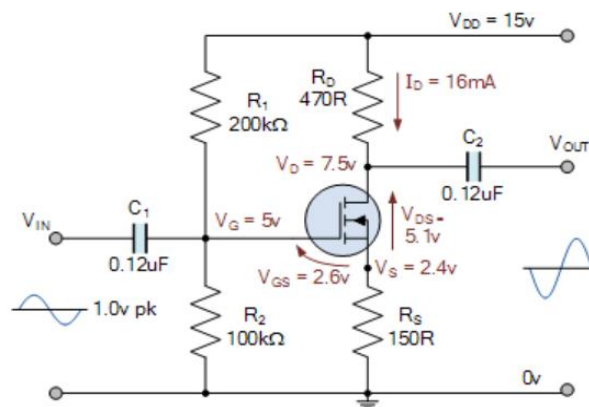
$$C_{ic} = \frac{10}{2\pi f_L(r_s + Z_i)} = \frac{10}{2\pi \times 200(0.6 \text{ k} + 667 \text{ k})} \approx 24 \text{ nF}$$

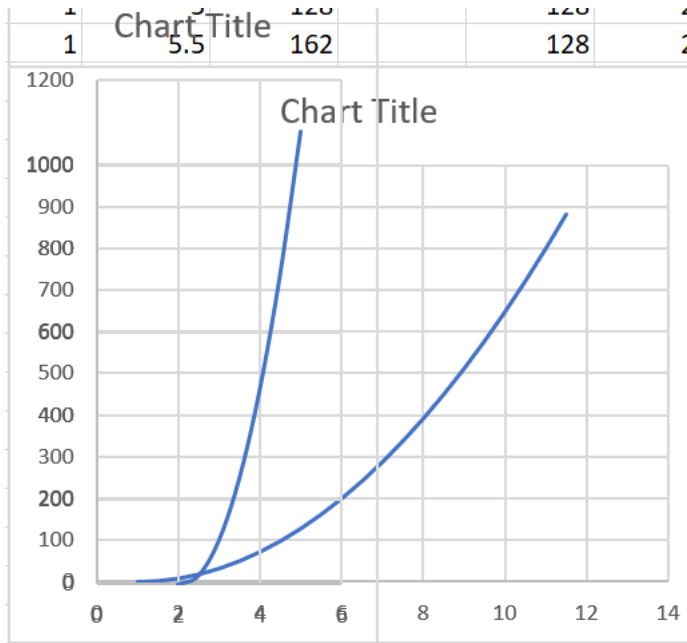
Output capacitor:

Using Equ.(10-32),
$$C_{oc} = \frac{10}{2\pi f_L(Z_o + R_L)} = \frac{10}{2\pi \times 200(0.56\text{ k} + 1\text{ k})} = 10\text{ }\mu\text{F (use } 2.7\text{ }\mu\text{F)}$$

Results: The designed common-gate amplifier is shown in Fig.10-27.

Comment: As the current capacity of the DMOSFET is very large, here to get linearity large value of I_D has been considered. Due to this large I_D , the values of R_D and R_S have been so small.

Single Stage MOSFET Amplifier



Value of r_e
$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{176 \text{ mA}} \approx 0.15 \Omega$$

Using Equ.(10-23),
$$C_{ic} = \frac{1}{2\pi f_L [r_s + \beta(r_e + R_E \parallel R_L)]}$$

or,
$$C_{ic} = \frac{1}{2\pi \times 100 [600 + 100(0.15 + 45.5)]} \text{ F} = 0.31 \mu\text{F} \text{ (use } 1 \mu\text{F)}$$

Output coupling capacitor:

Using Equ.(10-25),
$$C_{oc} = \frac{10}{2\pi f_L (r_e \parallel R_E + R_L)}$$

or,
$$C_{oc} = \frac{1}{2\pi \times 100 (4.3 \parallel 500 + 100)} \text{ F} \approx 0.71 \mu\text{F} \text{ (use } 1 \mu\text{F)}$$

Results: The designed amplifier circuit is shown in Fig.10-12.

Comments: Due to low value of R_B , current gain of this amplifier will be about 14. To design high current gain CC amplifier, we can use transistor with high h_{fe} or Darlington pair transistor.

10.11 Single-Stage Amplifier Design with PNP Transistor

The design process of PNP transistor amplifiers is exactly same as the design process of with NPN transistor amplifiers. The design procedures discussed for CE, CB, and CC amplifiers can be used for CE, CB, and CC amplifiers with PNP transistors. Here, only the CE amplifier is discussed. Although the design considerations are same, the biasing process of PNP transistor is different. The biasing process of PNP transistor has been discussed in Chapter 4. The circuit diagram of a single-stage common emitter amplifier with PNP transistor is shown in Fig.10-13, and its design process is discussed in Example 10-.

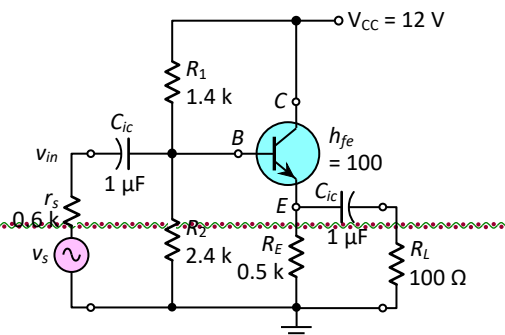


Fig.10-28: CC amplifier designed in Example 10-6

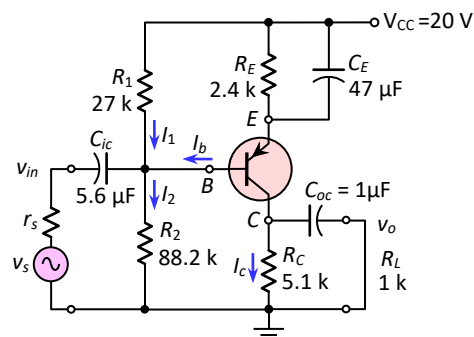


Fig.10-29: CE amplifier designed in Example 10-7

Example 9.7

Design a small signal CE amplifier using PNP transistor for the following specifications:

$V_{CC} = 20 \text{ V}$, $r_s = 600 \Omega$, $R_L = 3.3 \text{ k}$, lower cut-off frequency, $f_1 = 200 \text{ Hz}$, transistor $h_{fe} = 50$, $h_{ie} = 1 \text{ k}$.

10.12 Distortion in Amplifier

Distortion means the change in the shape of the signal waveform at the output of the system/circuit. In amplifiers, it means that the output waveform is different from the input waveform. If the input signal is a sine wave the output should also be a true sine wave. But in all the cases it may not be so, which we call **distortion**. Distortion can be due to the nonlinear characteristic of the device, due to operating point not setting properly, due to large signal swing of the input from the operating point or due to the reactive elements L and C in the circuit. Distortion is mainly classified as,

- a) Non-linear distortion
 - Amplitude distortion
- b) Linear distortion

- Frequency distortion and Phase distortion
- Phase - shift delay distortion

The classification tree is shown in Fig.10-30.

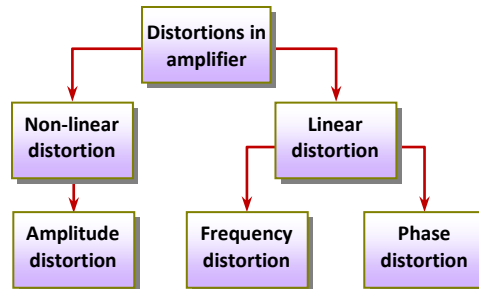


Fig.10-30: Classifications of

10.12.1 Non-linear distortion or Amplitude distortion

This is also called **amplitude distortion** or **harmonic distortion**. This type of distortion mainly occurs in large signal amplifiers or power amplifiers. It is due to the nonlinearity of the characteristic of the device.

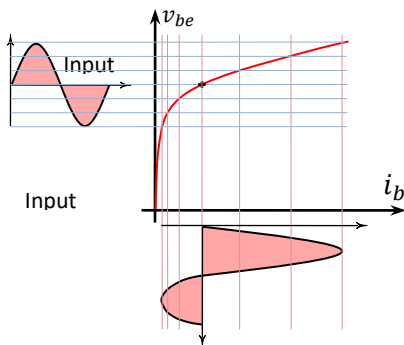
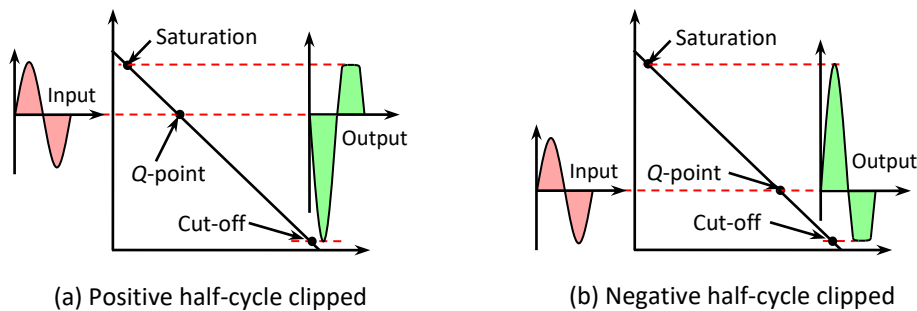


Fig.10-31: Common-base input characteristic (NPN)

Transistors and diodes are particularly **nonlinear**. Both BJTs and FETs used to design amplifiers. Depending upon operating point, the characteristics of these devices have usually exponential characteristics. This nonlinear characteristics produce distortion on the amplified output signal. The mechanism is illustrated in Fig.10-31. To make the concept clearer, the operating point is set at the lower level of the characteristics. Due to the nonlinear characteristics the positive half cycle has been amplified more than the negative half-cycle. The shape of the output signaled has been changed or distorted.

Amplitude distortion may, also, occur due to the incorrect setting of the *Q*-point when a large swing is produced. The situation is illustrated in Fig.10-32. If the *Q*-point is not set at the middle of the load-line, and the swing of the output is large, one half-cycle of the signal may be clipped-off.

Fig.10-32 **half-clip**: Amplitude distortion due incorrect position

This clipping of the one-half-cycle will introduce a serious distortion in the output signal.

10.12.2 Frequency distortion

Frequency Distortion is another type of amplifier distortion which occurs in amplifiers when the level of amplification varies with frequency. Many of the input signals that a practical amplifier amplifies consist of a **fundamental frequency** plus a number of different frequencies called **harmonics**. Normally, the amplitudes of these harmonics are a fraction of the fundamental amplitude and therefore have very little effect on the output waveform. However, the output waveform can become distorted if these harmonic frequencies are not amplified equally. The amplification will not be same for all harmonics. For example, consider the waveform shown in Fig.10-33. For simplicity, we have considered that the input signal has two components – the fundamental signal (500 Hz) and the second harmonics (1000 Hz). The resultant output waveform is shown on the right hand side. But when passed through the amplifier, the fundamental component has amplified by a factor 5, but the 2nd harmonics has not been amplified by 3. For this frequency dependant amplification, the output waveform has been changed (see Fig.(b) and Fi.(d)).

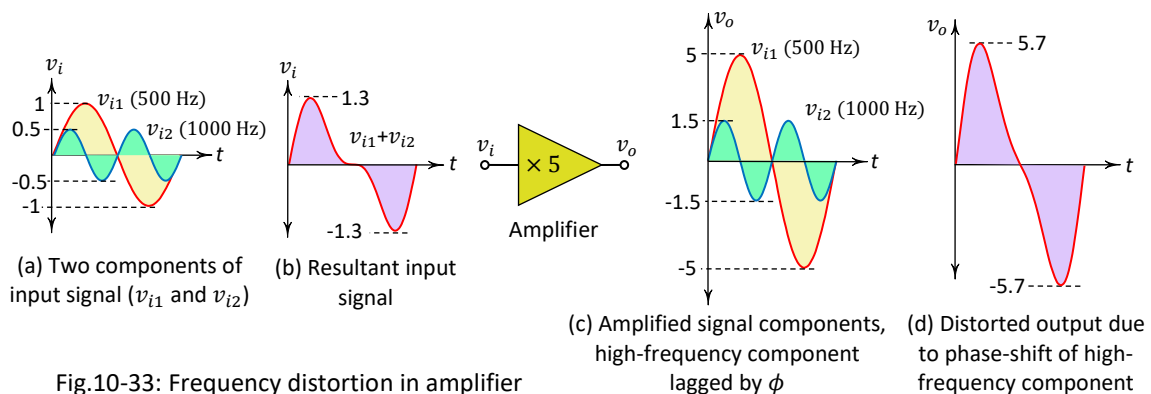


Fig.10-33: Frequency distortion in amplifier

10.12.3 Phase - Shift Delay Distortion

There will be phase shift between the input and the output and this phase shift will not be the same for all frequency signals. It also varies with the frequency of the input signal. In the output signal, all these distortions may be present or anyone may be present because of which the amplifier response will not be good. The situation is illustrated in Fig.10-33. The input signals are same as described in Fig.10-33. Here all the frequency components have been amplified equally, but the 2nd harmonic has been delayed by $\phi = 30^\circ$. Due to this delay (or lagging) the shape of the output signal has been changed as shown in Fig.10-33(d).

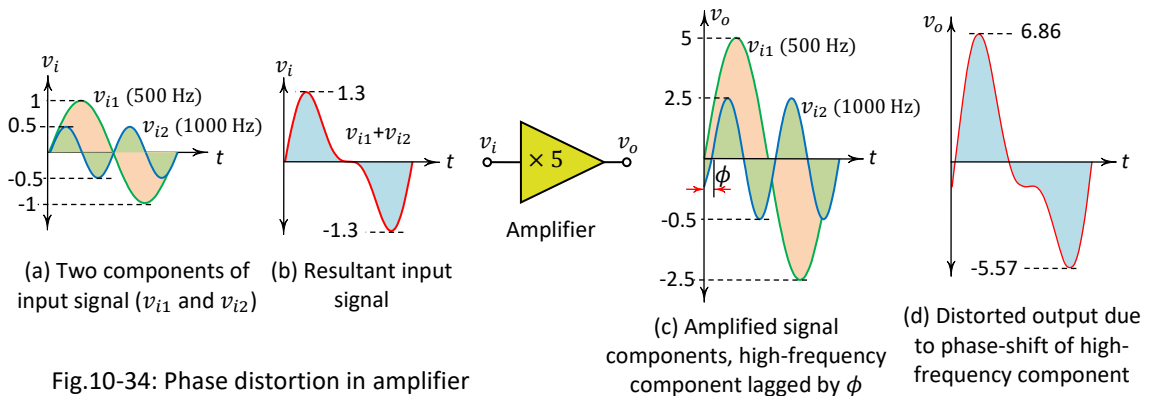


Fig.10-34: Phase distortion in amplifier

10.12.4 Reducing Distortion

One way to reduce distortion in Fig. 8-10 is by keeping the input signal small. When we reduce the peak value of the base voltage, we can reduce the movement of the instantaneous operating point. The smaller this swing or variation, the less the curvature in the characteristic graph. If the signal is small enough, the graph appears to be linear. Why is this important? Because there is negligible distortion for a small signals. When the signal is small, the changes in AC emitter current are almost directly proportional to the changes in AC base voltage because the graph is almost linear. In other words, if the ac base voltage is a small enough sine wave, the AC emitter current will also be a small sine wave with no noticeable stretching or compression of half-cycles.

10.13 Sample Questions

1. What do you mean by single-stage amplifier and small-signal amplifier? Explain with necessary figures.
2. Draw the circuit diagram of single-stage common-emitter (CE) amplifier and describe its operation.
3. Explain how CE amplifier produces 180° phase-shift of the output signal.
4. Draw the waveforms of the signal developed at different nodes of a CE amplifier. Explain why these shapes are found.
5. Explain how to design a single-stage common-emitter (CE) amplifier.
6. In a common-emitter amplifier which capacitor imposes the lower cut-off frequency? Explain why.

7. Derive the expression to calculate the input coupling capacitor (C_{ic}) of the CE amplifier for a given lower cut-off frequency.
8. Derive the expression to calculate the output coupling capacitor (C_{oc}) of the CE amplifier for a given lower cut-off frequency.
9. Derive the expression to calculate the emitter bypass capacitor (C_E) of the CE amplifier for a given lower cut-off frequency.
10. What are the advantages and disadvantages of a common-emitter amplifier over the other two amplifiers? What are the use of this amplifier?
11. Draw the circuit diagram of a single-stage common-base (CB) amplifier and describe its operation.
12. Explain why the phase-shift of a CB amplifier is 0° .
13. Draw the waveforms of the signals developed at different nodes of a CB amplifier. Explain why these shapes are found.
14. Explain how to design a single-stage common-base (CB) amplifier.
15. In a common-base amplifier which capacitor imposes the lower cut-off frequency? Explain why.
16. Derive the expression to calculate the input coupling capacitor (C_{ic}) of the common-base (CB) amplifier for a given lower cut-off frequency.
17. Derive the expression to calculate the output coupling capacitor (C_{oc}) of the CB amplifier for a given lower cut-off frequency.
18. Derive the expression to calculate the emitter bypass capacitor (C_E) of the CB amplifier for a given lower cut-off frequency.
19. What are the advantages and disadvantages of a common-emitter amplifier over the other two amplifiers? What are the use of this amplifier?
20. Draw the circuit diagram of a single-stage common-collector (CC) amplifier and describe its operation.
21. Explain why the phase-shift of a CC amplifier is 0° .
22. Draw the waveforms of the signals developed at different nodes of a CC amplifier. Explain why these shapes are found.
23. Explain how to design a single-stage common-collector (CC) amplifier.
24. Derive the expression to calculate the input coupling capacitor (C_{ic}) of the common-collector (CB) amplifier for a given lower cut-off frequency.
25. Derive the expression to calculate the output coupling capacitor (C_{oc}) of the CC amplifier for a given lower cut-off frequency.
26. Derive the expression to calculate the emitter bypass capacitor (C_E) of the CB amplifier for a given lower cut-off frequency.
27. In a common-collector amplifier which capacitor imposes the lower cut-off frequency? Explain why.
28. Explain what will happen if all the three capacitors of an amplifier impose the lower cut-off frequency?
29. What are the advantages and disadvantages of a common-collector amplifier over the other two amplifiers? What are the use of this amplifier?
30. Explain how to design a single-stage common-emitter (CE) amplifier with PNP transistor.
31. Explain how to design a single-stage common-base (CB) amplifier using PNP transistor.
32. Explain how to design a single-stage common-collector (CC) amplifier using PNP transistor.

33. Draw the circuit diagram of single-stage common-source (CS) amplifier and describe its operation.
34. Explain how CS amplifier produces 180° phase-shift at the output.
35. Draw the waveforms of the signals developed at different terminals of a CS amplifier. Explain why these shapes are found.
36. Explain how to design a single-stage common-source (CS) amplifier.
37. In a common-source amplifier which capacitor sets the lower cut-off frequency? Explain why.
38. Derive the expression to calculate the input coupling capacitor (C_{ic}) of a CS amplifier for a given lower cut-off frequency.
39. Derive the expression to calculate the output coupling capacitor (C_{oc}) of a CS amplifier for a particular lower cut-off frequency.
40. Derive the expression to calculate the emitter bypass capacitor (C_E) of the CS amplifier for a given lower cut-off frequency.
41. What are the advantages and disadvantages of a common-source (CS) amplifier over the other two amplifiers? What are the use of this amplifier?
42. Draw the circuit diagram of single-stage common-drain (CD) amplifier and describe its operation.
43. Explain why there is no phase-shift in CD amplifier.
44. Explain how to design a single-stage common-drain (CD) amplifier.
45. In a common-drain amplifier which capacitor sets the lower cut-off frequency? Explain why.
46. Derive the expression to calculate the input coupling capacitor (C_{ic}) of a CD amplifier for a given lower cut-off frequency.
47. Derive the expression to calculate the output coupling capacitor (C_{oc}) of a CD amplifier for a particular lower cut-off frequency.
48. Derive the expression to calculate the emitter bypass capacitor (C_E) of the CD amplifier for a given lower cut-off frequency.
49. What are the advantages and disadvantages of a common-drain amplifier over the other two amplifiers? What are the use of this amplifier?
50. Draw the circuit diagram of single-stage common-gate (CG) amplifier and describe its operation.
51. Explain why there is no phase-shift in CG amplifier.
52. Explain how to design a single-stage common-gate (CG) amplifier.
53. In a common-gate amplifier which capacitor sets the lower cut-off frequency? Explain why.
54. Derive the expression to calculate the input coupling capacitor (C_{ic}) of a CG amplifier for a given lower cut-off frequency.
55. Derive the expression to calculate the output coupling capacitor (C_{oc}) of a CG amplifier for a particular lower cut-off frequency.
56. Derive the expression to calculate the emitter bypass capacitor (C_E) of the CG amplifier for a given lower cut-off frequency.
57. What are the advantages and disadvantages of a common-gate amplifier over the other two amplifiers? What are the use of this amplifier?

10.14 Exercise

1. Design a small-signal common-emitter CE amplifier for the following specifications: $V_{CC} = 12\text{ V}$, $r_s = 600\ \Omega$, $R_L = 5.6\text{ k}$, lower cutoff frequency, $f_L = 100\text{ Hz}$, transistor $h_{fe} = 75$, and $h_{ie} = 1.2\text{ k}$.
2. Design a single stage CE amplifier with voltage divider bias that will give an open-circuit voltage gain of -50 . Assume, $V_{CC} = 20\text{ V}$, $I_{CQ} = 1.2\text{ mA}$, $f_1 = 100\text{ Hz}$, and $\beta_{min} = 120$.
3. Design a small-signal CB amplifier for the following given specifications: $V_{CC} = 16\text{ V}$, $r_s = 600\ \Omega$, $R_L = 5.1\text{ k}$, lower cut-off frequency, $L = 150\text{ Hz}$, transistor $h_{fe} = 110$.
4. Design a small-signal CC amplifier for the following given specifications: $V_{CC} = 16\text{ V}$, $r_s = 600\ \Omega$, $R_L = 50\ \Omega$, lower cut-off frequency, $f_L = 150\text{ Hz}$, minimum current gain of 60, and transistor $h_{fe} = 80$.
5. Design a small-signal CC amplifier for the following given specifications: $V_{CC} = 20\text{ V}$, $r_s = 200\ \Omega$, $R_L = 100\ \Omega$, $f_L = 200\text{ Hz}$, minimum current gain of 50, and transistor $h_{fe} = 100$.
6. Design a small-signal CC amplifier for the following given specifications: $R_L = 60\ \Omega$, lower cut-off frequency, $f_1 = 120\text{ Hz}$, minimum RMS current through the load is 100 mA, and $h_{fe} = 140$.
7. Design a small signal CE amplifier using PNP transistor for the following specifications: $V_{CC} = 22\text{ V}$, $r_s = 400\ \Omega$, $R_L = 5.1\text{ k}$, lower cut-off frequency, $f_1 = 150\text{ Hz}$, transistor $h_{fe} = 75$, $h_{ie} = 0.75\text{ k}$.
8. Suppose a JFET has $V_{P(\max)} = -6\text{ V}$, and $V_{P(\min)} = -3\text{ V}$, $I_{DSS(\max)} = 20\text{ mA}$ and $I_{DSS(\min)} = 10\text{ mA}$ and is given in Fig.10-16. Design a CS amplifier considering $V_{DD} = 22\text{ V}$, lower cut-off frequency = 250 Hz, $R_L = 6.2\text{ k}$ and $r_s = 600\ \Omega$.

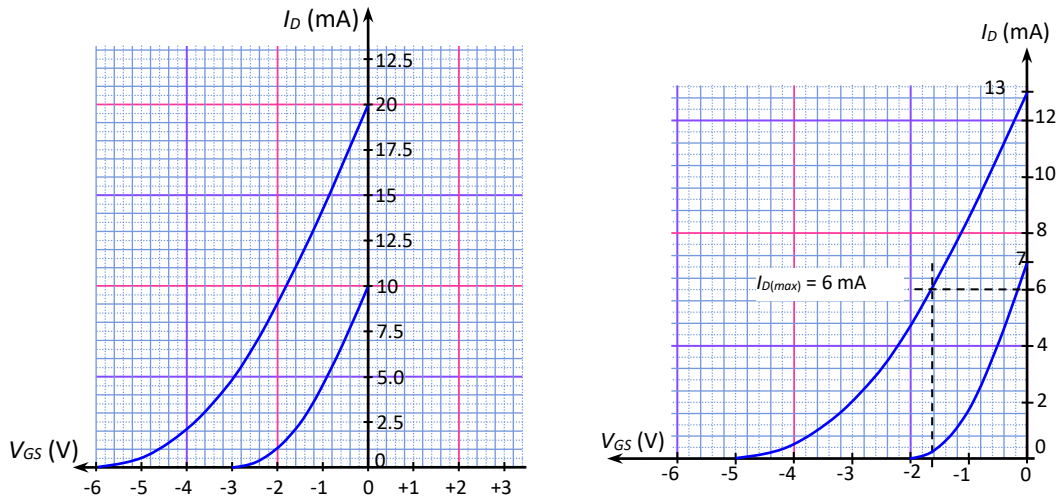


Fig.10-16: Transfer characteristics and bias current for Example 9.8

9. Suppose a JFET has $V_{P(\max)} = -5\text{ V}$ and $V_{P(\min)} = -2.5\text{ V}$, and $I_{DSS(\max)} = 15\text{ mA}$ and $I_{DSS(\min)} = 7.5\text{ mA}$. Design a CD amplifier considering $V_{DD} = 22\text{ V}$, lower cut-off frequency =

100 Hz, $R_L = 5.6 \text{ k}$ and $r_s = 250 \Omega$. [You may use the transfer characteristics of Fig.10-16 by changing the labeling of the x and y -axis.]

10. JFET 2N5951 has $V_{P(\text{max})} = -5 \text{ V}$ and $V_{P(\text{min})} = -2 \text{ V}$, and $I_{DSS(\text{max})} = 13 \text{ mA}$ and $I_{DSS(\text{min})} = 7 \text{ mA}$. Design a CG amplifier considering $V_{DD} = 20 \text{ V}$ and lower cut-off frequency = 150 Hz, $R_L = 4.3 \text{ k}$ and $r_s = 500 \Omega$. The minimum gain of the amplifier should be 7.5. Use the transfer characteristics of Fig.10-22.

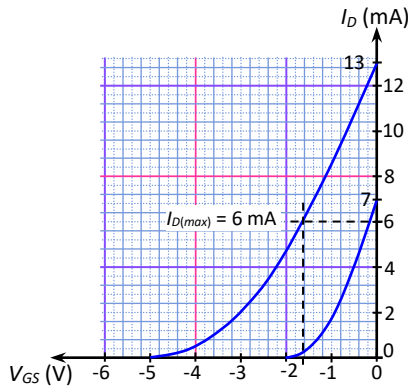


Fig.10-22: Transfer characteristics and bias current for Example 9.10

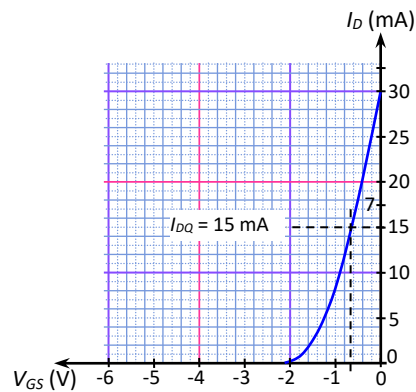


Fig.10-24: Transfer characteristics and bias current for Example 9.11

11. BSS139 is an N-channel DMOSFET with $V_P = -2.1 \text{ V}$, $I_{DSS} = 30 \text{ mA}$, and $R_{GS} = 10^9 \Omega$. Design a CG amplifier considering $V_{DD} = 25 \text{ V}$, $f_L = 200 \text{ Hz}$, $R_L = 2.2 \text{ k}$ and $r_s = 600 \Omega$. The amplifier must be capable to produce an AC voltage swing of $\pm 2.2 \text{ V}$ minimum. Use the transfer characteristics of Fig.10-24.
12. 2N7002-7-F is a small signal EMOFET. The minimum and the maximum transfer characteristics are given in Fig.10-26. Design a CS amplifier considering $f_L = 200 \text{ Hz}$, $R_L = 3 \text{ k}$ and $r_s = 600 \Omega$. The minimum gain of the amplifier should be 50 and the amplifier should produce an AC swing of $\pm 2.2 \text{ V}$ at the output. Suggest the minimum value of V_{DD} suitable for this amplifier.

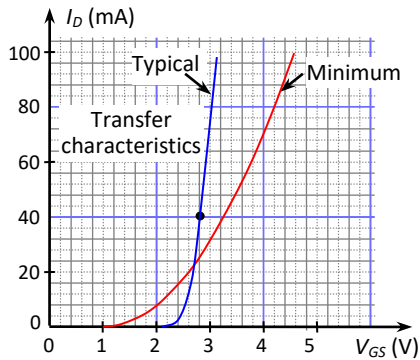


Fig. 9.26: Transfer characteristics for Exercise 12

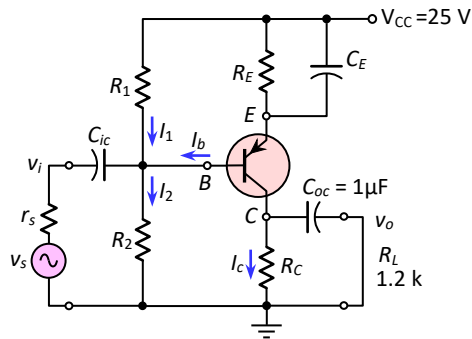


Fig.10-29: CE amplifier designed in Exercise 13

13. A Design a small signal CE amplifier as given in Fig.10-29. Assume, $V_{CC} = 25 \text{ V}$, $r_s = 600 \Omega$, $R_L = 5.1 \text{ k}$, lower cut-off frequency, $f_1 = 250 \text{ Hz}$, transistor $h_{fe} = 75$, $h_{ie} = 1.2 \text{ k}$.